
Measurement of WbWb and more

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Munich, Hoffentlich bald

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Acknowledgments

Abstract

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1 Introduction

Top-quark pair ($t\bar{t}$) production is one of the most studied processes at the Large Hadron Collider (LHC) and the measurement of differential cross sections is of fundamental importance, since $t\bar{t}$ production provides access to fundamental parameters of the Standard Model (SM) of particle physics, like the top-quark mass, but it is also an important background for many other measurements and searches for physics Beyond the Standard Model (BSM).

The top-quark pair decays predominantly into two W bosons and two b -quarks ($WbWb$). Differential cross sections for $t\bar{t}$ production have been measured [1–5] and calculated [6–8] in a wide kinematic range. All these measurements consider the $WbWb$ final state.

Further processes contribute to the $WbWb$ final state, most prominently the associated single-resonant top-quark production process tWb (shown on a Feynman diagram in ??). These contributions are considered as background processes in top-quark pair production analyses, but these NLO real corrections to the LO tW production process interfere with $t\bar{t}$ and with each other and therefore cannot be unambiguously treated theoretically. The interference between $t\bar{t}$ and tW therefore reduces the achievable precision for top-quark phenomenology using top-quark pair cross-sections. In contrast, the direct measurement of $WbWb$ production cross-sections provides an alternative access to top-quark phenomenology, where those aspects are treated consistently in theory, albeit at the cost of considerably more complicated calculations. The interference between top-quark pair production and single-top production at higher-order perturbative QCD also introduces a considerable uncertainty in searches for new physics phenomena at large masses, and this measurement aims to reduce this uncertainty for future searches.

This interference problem has been widely studied from the theoretical point of view [9–13] and provides a significant source of uncertainty for many measurements [2, 14, 15] and BSM searches [16–21]. The consistent calculation of the $WbWb$ final state was performed up to next-to-leading order in QCD [22], including full top off-shell effects at NLO, and supplemented with parton showers [23, 24]. A calculation of the process $pp \rightarrow \ell v b b j j$ was performed up to NLO in Ref. [25]. The only existing measurement available to assess these predictions in a region sensitive to interference is the one provided by ATLAS in 2018 [26] in the dilepton channel.

In this note, we expand on that result to cover also semileptonic $WbWb$ decays yielding a final state with a charged lepton, significant missing transverse momentum, two b -jets and light jets. When compared to the dilepton channel, the semileptonic decay channel provides the opportunity to measure the W boson kinematics directly, since the hadronically decaying W boson can be fully reconstructed, and in addition, the larger

cross section enables us to probe a larger phase space up to higher scales, which is important to constrain the $t\bar{t}$ and tW interference effects at high scales, which is relevant for BSM searches.

The analysis therefore focusses on three kinematic regions: a large phase space to constrain the interference effects and improve the modelling over a large kinematic region; an analysis region targeting the interference in phase-spaces typically addressed by BSM physics searches; and a dedicated measurement region with a focus on the W boson and b -jet kinematic observables. Differential cross-section measurements are performed for selected observables relevant for $t\bar{t}$ physics. These differential cross-sections can be used to assess and tune the performance of new Monte Carlo generators, determine SM parameters, or constrain uncertainties in BSM searches. The measurements are required to ensure that basic kinematics in the event are well modelled when the interference is correctly modelled. The full dataset collected by the ATLAS detector between 2015 and 2018 at $\sqrt{s} = 13$ TeV center-of-mass energy and $\mathcal{L} = 140 \text{ fb}^{-1}$ is used for the measurement.

2 Theoretical Framework

Some theoretical basics.

2.1 The Standard Model of Particle Physics

Describe the SM.

2.2 Monte Carlo Event Generators

Pythia, Herwig and Sherpa?

3 The ATLAS experiment at the Large Hadron Collider

The methods and techniques in experimental particle physics have undergone a remarkable transformation. From the discovery of the positron [27] and the muon [28] in cloud chambers in the 1930s, over the discovery of neutral weak currents at the Gargamelle bubble chamber [29] to the discovery of the Top quark at Fermilab [30, 31] and the Higgs Boson at the Large Hadron Collider (LHC) [32, 33].

The LHC is the largest and most powerful accelerator built to this day. It is hosted by the European Organization for Nuclear Research (CERN) in Geneva, Switzerland. This chapter gives an introduction to the CERN accelerator complex, including the LHC, before discussing one of its major experiments, A Toroidal LHC ApparatuS, better known as ATLAS.

3.1 The Large Hadron Collider

3.1.1 The CERN accelerator complex

The accelerator complex at CERN is a succession of machines that accelerate protons to increasingly higher energies. Each machine boosts the energy of a beam of particles before injecting it into the next machine in the sequence. The layout of the complex is shown in Figure 3.1.

The proton source is a bottle of hydrogen gas. Protons are obtained by stripping the hydrogen atoms of their electrons with an electric field. The Linear accelerator 2 (Linac2) is the first module, accelerating protons to an energy of 50 MeV before injecting them to the Proton Synchrotron Booster (PSB).

After the acceleration to 2 GeV by the PSB, they are subsequently pushed to 26 GeV and 450 GeV by the Proton Synchrotron (PS) and Super Proton Synchrotron (SPS), respectively.

The final stage of the accelerator complex is the LHC [35], a two-ring-superconducting-hadron accelerator and collider. It was installed in the existing tunnel that was constructed for the Large Electron Positron (LEP) [36] machine operating from 1989 to 2000. It lies around 100 m beneath the surface and has a circumference of 26.7 km.

The tunnel has an octagonal structure consisting of eight straight sections connected by arcs. The LHC ring accommodates 1232 dipole magnets consisting of niobium-titanium coils. The magnets are cooled below 2 K with superfluid helium to operate at magnetic fields above 8 T. To account for the space limitations and reduce construction cost, a

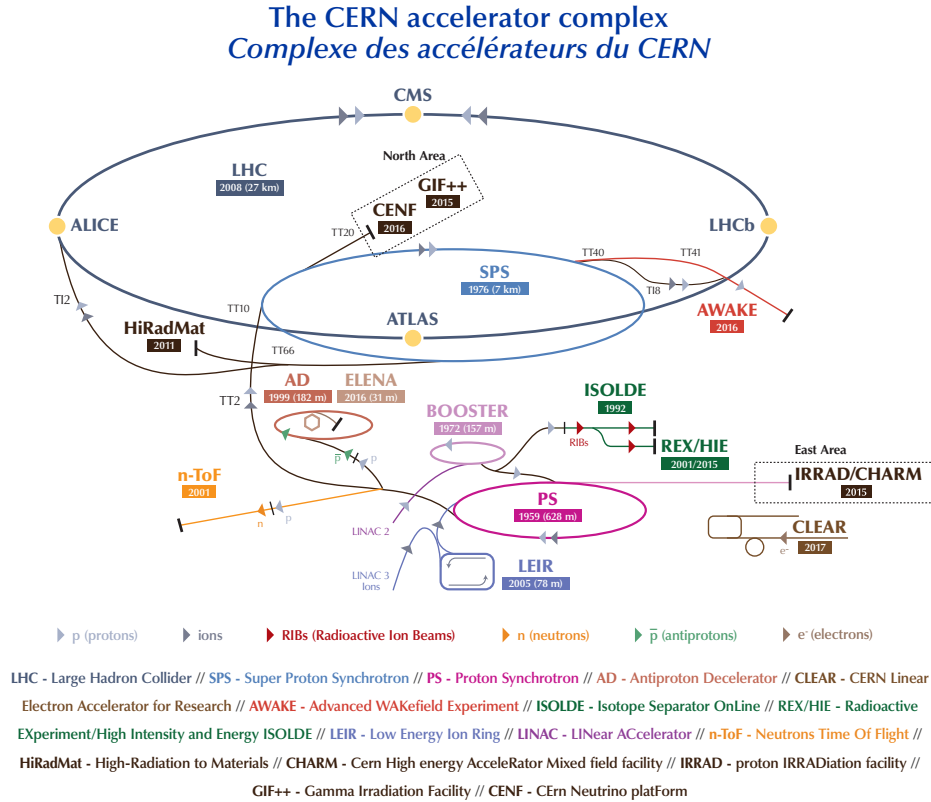


Figure 3.1: Overview of CERN accelerator complex showing pre-accelerators and the LCH (dark blue) with the four major experiments ATLAS, CMS, ALICE and LHCb [34].

"two-bore" system is adopted. This two-in-one design accommodates the windings for the two beam channels in a common cold mass and cryostat.

It takes around five minutes to fill each LHC ring with proton bunches with a spacing of 25 ns. Within 20 minutes the protons are accelerated to reach their maximum energy of 6.5 TeV. Beams circulate for many hours inside the LHC beam pipes under normal operating conditions [37].

The acceleration chain can also be used for heavy ions, such as lead, to complement and extend the proton physics program. They are fed in the Linear accelerator 3 (Linac3) before further acceleration and storage in the Low Energy Ion Ring (LEIR). Afterwards they follow the same route as the protons, reaching their maximum energy in the LHC. Since heavy ion physics is not relevant in the scope of this thesis, it will not be discussed further.

The two beams are brought into collision at four designated locations along the ring, the Interaction Points (IP). Four major experiments are positioned at these locations. ATLAS [38] and CMS [39] are two general purpose detectors which study a broad range

of physics topics from precise measurements of the SM, including top quark and Higgs boson, to searches for BSM physics. They are complemented by two more specialized detectors. ALICE [40] is dedicated to heavy-ion physics and LHCb [41] focuses on heavy-flavor physics.

3.1.2 Operation

The LHC operation is divided into distinct data taking periods, so-called *Runs*, and intermediate *Long Shutdowns* (LS). LHC operation started in 2009 with Run 1. The center of mass energy was gradually increased from 900 GeV to 7 TeV and later 8 TeV, before Run 1 concluded in 2012. Upgrades to the magnet system were performed during LS 1. This allowed the LHC to operate at $\sqrt{s} = 13$ TeV during Run 2 from 2015 to 2018. Further improvements during LS2 until 2022 allowed data taking at $\sqrt{s} = 13.6$ TeV during Run 3, which is scheduled to continue until 2026.

During the next planned LS, the LHC will be substantially upgraded to the High-Luminosity LHC (HL-LHC) [42], targeting collisions rate five times larger than the initial design of the LHC.

A *fill* denotes a single uninterrupted period with beams circulating in the LHC. Once the LHC reaches conditions indicating stable particle collisions in the IP, ATLAS ramps up the high voltage (HV) in several detector subsystems. This marks the start of continuous data recording.

3.1.3 Luminosity and pile-up

The instantaneous luminosity $\mathcal{L}_{\text{inst}}$ is one of the most important parameters of any accelerator. Together with the cross section σ_X , which represents the probability of a process X to occur upon collision, it defines the rate of collisions

$$\frac{dN_X}{dt} = \mathcal{L}_{\text{inst}} \cdot \sigma_X. \quad (3.1)$$

The luminosity is measured in *barn* (b), where 1b corresponds to $10^{-24}\text{cm}^{-2}\text{s}^{-1}$. It depends on the beam parameters and is given by

$$\mathcal{L}_{\text{inst}} = \frac{N_1 N_2 f N_b}{4\pi\sigma_x\sigma_y}, \quad (3.2)$$

with the number of protons per bunch N_1, N_2 , the frequency of collisions f determined by the revolution frequency of the LHC, the number of bunches in each beam N_b and the horizontal and vertical bunch dimensions σ_x and σ_y . The LHC was designed for a peak instantaneous luminosity of $10^{-34}\text{cm}^{-2}\text{s}^{-1}$ for pp collisions [43]. This goal was superseded during Run 2 by almost a factor of two [44].

A measure for the number of events N is related to the integrated luminosity L which can be obtained by integrating $\mathcal{L}_{\text{inst}}$ over a given period

$$N = \sigma \cdot L = \sigma \int \mathcal{L}_{\text{inst}}(t) dt. \quad (3.3)$$

The number of protons per bunch is of the order 10^{11} . The collision frequency is 40 MHz during normal operation. Therefore, it is common that multiple pp collision occur at the same time in a single bunch crossing (*in-time*) as well as over bunch crossings (*out-of-time*). This leads to overlapping detector signals in addition to the primary collision of interest. This phenomenon is usually referred to as *pile-up* μ . The μ distribution is shown in Figure 3.2(a) for the four years of data taking in Run 2. The mean pile-up for each year of data taking ranges from 13.4 to 37.8.

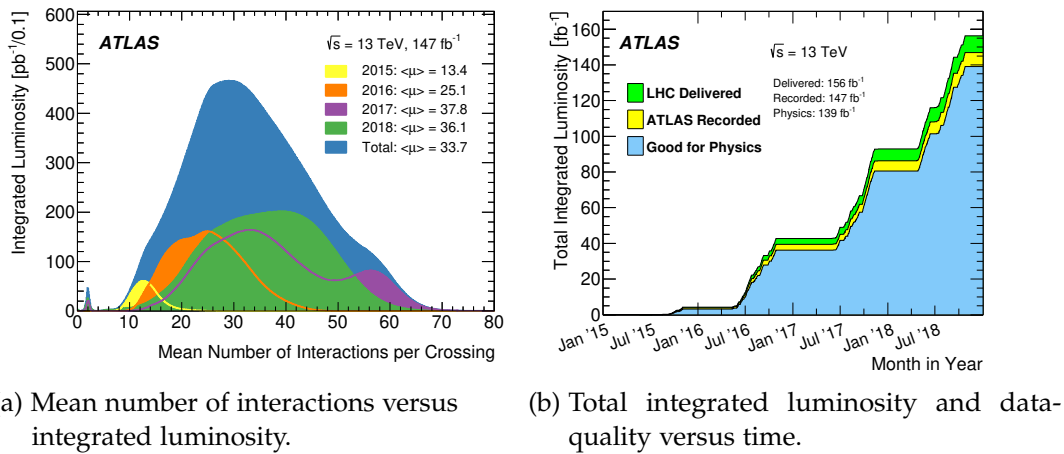


Figure 3.2: Plots summarizing luminosity and data quality reported by the LHC and ATLAS during Run 2 from 2015 to 2018. 3.2(a) Distribution of the mean number of interactions per bunch crossing, μ , weighted by the luminosity. The individual distributions for each year of pp collision data taking during Run 2 are shown. The mean μ value for each year are given in the legend. 3.2(b) Cumulative integrated luminosity delivered to and recorded by the ATLAS experiment during Run 2. Data recorded at $\sqrt{s} = 13$ TeV during stable beam pp collisions is considered, including special runs for detector calibration. The data sample certified for physics analysis usage is shown in light blue with the *Good for Physics* label. Figures taken from Ref. [45].

The integrated luminosity collected by the ATLAS experiment is shown in Figure 3.2(b). It is slightly lower than the luminosity delivered by the collider due to detector inefficiencies. The data that can be used for physics analyses is called *Good for Physics*. The full Run 2 data sample suitable for physics analyses corresponds to an integrated luminosity of $140.1 \pm 1.2 \text{ fb}^{-1}$, after applying data-quality selections [44].

To put things in perspective, when using the $t\bar{t}$ production cross section of 829 pb [46], the number of $t\bar{t}$ pairs produced during Run 2 amounts to approximately $1.3 \cdot 10^8$.

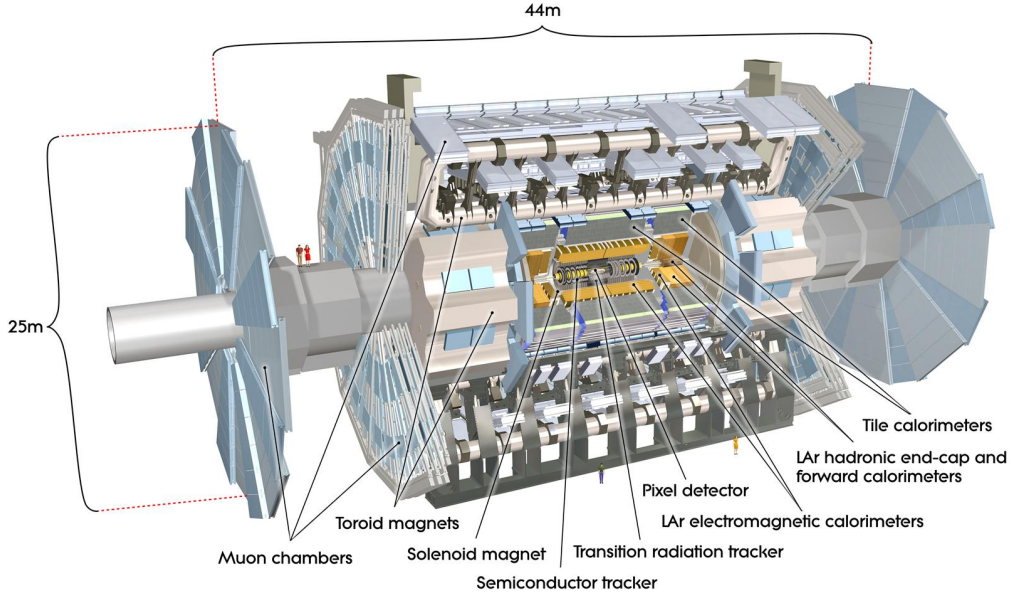


Figure 3.3: Schematic diagram of the ATLAS detector [47].

3.2 The ATLAS detector

The data analyzed in the scope of this thesis was collected by the ATLAS experiment [38]. This section provides a short summary of the detector subsystems relevant for this measurement, as well as an overview of the data triggering and processing stream. Since the discussed data was taken during Run 2, the Run 2 detector configuration will be discussed. A schematic drawing of the machine is provided in Figure 3.3. With a length of 44 m and a diameter of 25 m, ATLAS is the largest of the four LHC experiments.

3.2.1 Coordinate system

A right-handed Cartesian coordinate system centered at the IP is employed in ATLAS, as illustrated in Figure 3.4. The z -axis aligns with the beam line, the x -axis points towards the center of the LHC ring and the y -axis is oriented vertically upwards. Since the LHC collides partons from the protons, the initial momentum along the z -axis is unknown. Observables are often defined in the transverse plane, spanned by the x - and y -axes. An example is the transverse momentum

$$p_T = \sqrt{p_x^2 + p_y^2}. \quad (3.4)$$

Instead of Cartesian coordinates, the use of spherical coordinates is more common in particle physics. The 3-vectors of particles are defined by the magnitude r , polar angle θ defined from the beam axis and the azimuthal angle φ around the beam axis. The

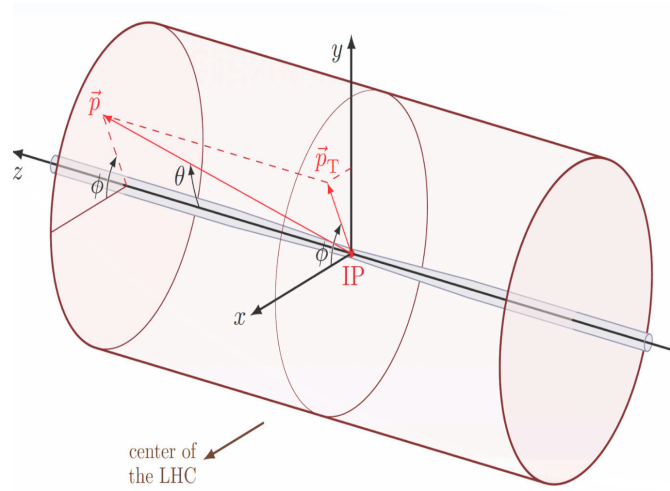


Figure 3.4: Illustration of the ATLAS coordinate system. Figure taken from Ref. [48].

polar angle ranges from 0 to π , where $\vartheta = 0$ points in the $+z$ -direction. The detector is designed to be symmetrical in φ . For a particle with Energy E and z component of the momentum p_z , the rapidity y can be defined as

$$y = \frac{1}{2} \ln \left(\frac{E + p_z}{E - p_z} \right). \quad (3.5)$$

This proves to be a sensible definition, since differences in y between particles from the same collision are invariant under Lorentz transformation along the z -axis. In the limit where particle masses can be neglected, E is equivalent to the magnitude of the 3-momentum vector $p = |\vec{p}|$. The expression can be simplified to obtain the *pseudo-rapidity*

$$\eta = \ln \left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z} \right) = -\ln \tan \left(\frac{\vartheta}{2} \right). \quad (3.6)$$

3.2.2 Magnet system

The ATLAS magnet system [49] is build out of four large superconducting magnets: the central solenoid, the barrel toroid and two end-cap toroids. It provides a magnetic field strength exceeding 50 mT over a volume of approximately 12000 m³. The magnets are necessary for particle charge identification and allow precise momentum measurements within the tracking and muon systems.

Figure 3.5 shows how the three toroids are embedded in the machine. The eight coils of the barrel toroid and the two barrel toroids produce a magnetic field of 0.5 T and 1.0 T for the central and end-cap muon detectors, respectively.

A complementary solenoid is embedded inside the Liquid Argon (LAr) Calorimeter. It is aligned with the beam axis and provides a 2 T magnetic field immersing the inner

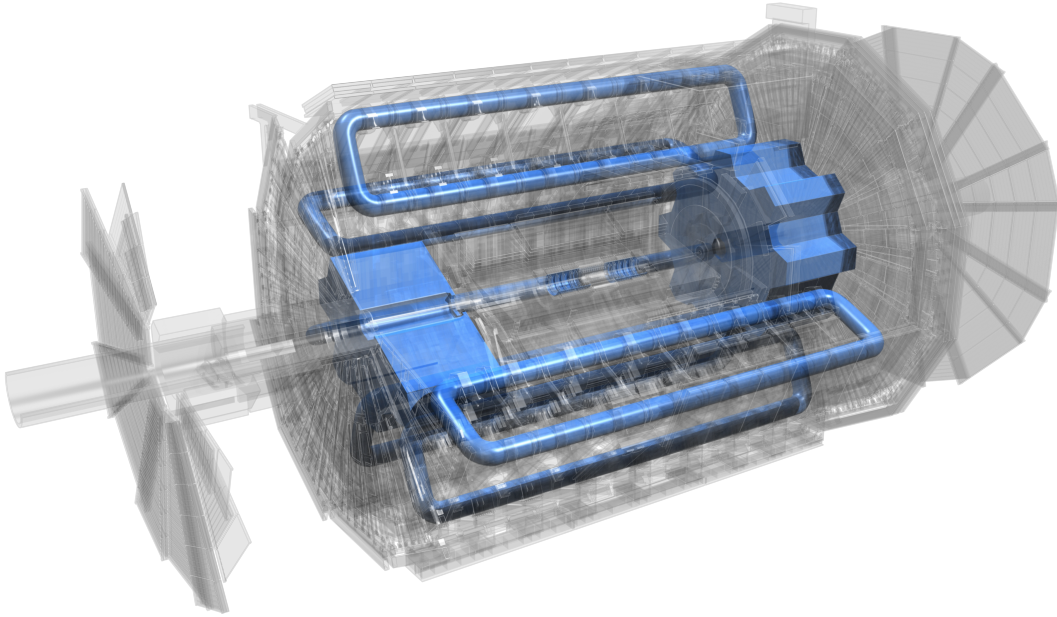


Figure 3.5: Geometry of the ATLAS magnet system. The eight coils of the barrel toroid and the two end-cap toroids are highlighted in blue. The solenoid winding which is located in the calorimeter volume is not shown [50].

detector. The solenoid is designed to minimize the radiative thickness in front of the barrel calorimeter by sharing a common vacuum vessel with the LAr calorimeter.

3.2.3 Inner detector

Closest to the beam line is the ATLAS Inner Detector (ID) [51, 52], consisting of three independent but complementary subdetectors. It measures the electrical charge and momentum of charged particles within $|\eta| < 2.5$. It provides excellent momentum resolution for the reconstruction of primary and secondary vertices. A schematic labeled drawings of the ID layout is shown in Figure 3.6. It shows the silicon based Pixel and Semiconductor Tracker (SCT), surrounded by the gas based Transition Radiation Tracker (TRT).

In order to suppress noise to an adequate level, the silicon sensors are cooled to temperatures around 0°C or slightly below. No cooling is required for the TRT [38]. The solenoid provides a 2 T axial magnetic field for the entire ID system. The individual subsystems are briefly introduced in the following.

Pixel detector

The innermost part of the ID is the ATLAS pixel detector [54]. In the original design, it consisted of three layers in the barrel (B-layers) and three disks on each side of the

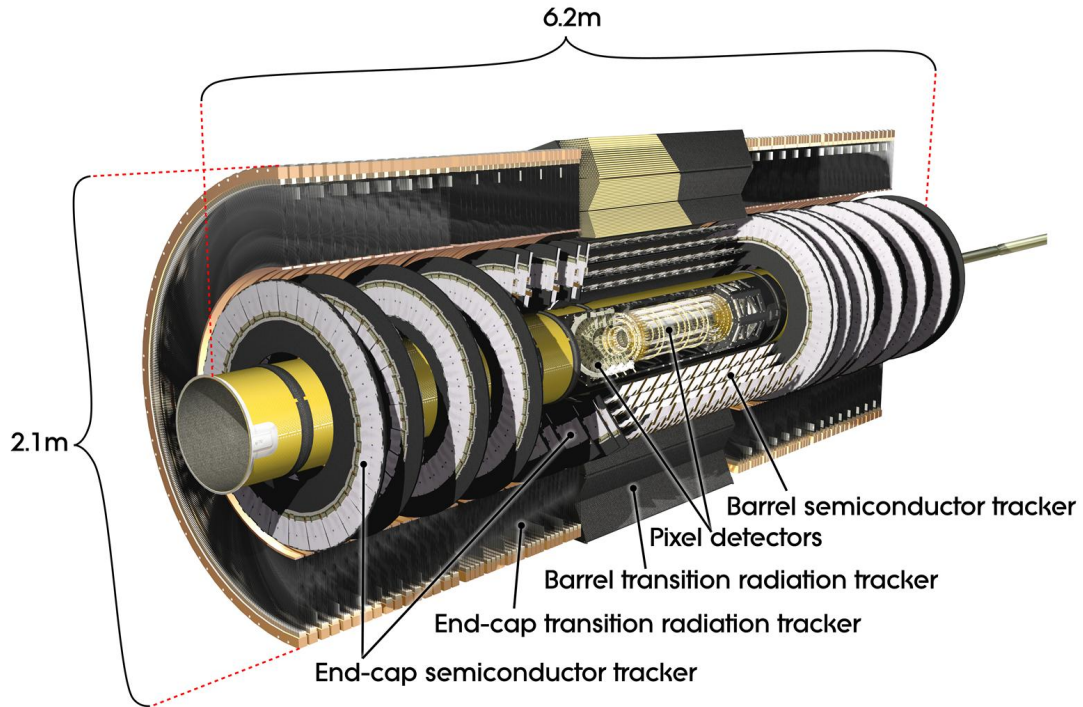


Figure 3.6: Labeled drawing of the ATLAS Inner Detector [53].

interaction region During LS1, a new smaller radius beam pipe was installed, allowing the installation of an additional layer, the Insertable B-Layer (IBL)[55, 56]. The mean sensor radius from the IP was reduced from 50.5 mm to 33.5 mm. The lifespan of the components due to irradiation could be prolonged.

The ID resolution is mainly driven by the cell size. It is $500 \times 50 \mu\text{m}^2$ in the original design for the three outer layers and could be reduced to $250 \times 50 \mu\text{m}^2$ for the IBL sensors. The ID provides full azimuthal coverage for particles exceeding $p_T > 1 \text{ GeV}$ and three or more space points over the acceptance of the ID.

When a charged particle traverses a silicon sensor, it creates electron-hole pairs in the doped silicon substrate. An applied electric field forces the pairs to drift to opposite sides of the sensor, generating an electrical current. The amount of collected charge is related to the energy the particle lost, while the spatial coordinates are given by the pixels position.

Semiconductor tracker

The SCT [57] also employs silicon as active detector material with the same working principle as in the pixel detector. However, instead of pixels, strips with a width of $80 \mu\text{m}$ and length of 6 cm are used (12 cm for the two outermost strips). The SCT consists of 4088 silicon strips, extending the tracking system to a radial distance of

$299 < r < 560$ mm. The strips are arranged in four layers in the barrel and nine disks in each of the endcaps.

Using strips instead of pixels lowered the manufacturing cost at the prize of reduced granularity. Two planes of silicon sensors are glued together at an angle of 40 mrad to improve the spatial resolution.

Transition Radiation Tracker

The TRT [58] is the outermost part of the ID. Its building blocks are thin-walled proportional drift tubes, also called straw tubes due to its small diameter of only 4 mm. The tubes are made from multilayer Kapton films and are reinforced with carbon fibre bundles. Each of the tubes has a gold-plated tungsten wire in the center and is filled with an argon-based gas mixture. The TRT consists of a barrel region with 144 cm long oriented parallel to the beam line, as well as endcaps with 37 cm long tubes arranged radially.

A charged particle crossing the TRT ionizes gas which results in an electrical signal in the single tubes. Additionally, transition radiation from the tubes can be used for particle identification. A particle with $p_T > 0.5$ GeV and $|\eta| < 2.0$ creates a signal in around 30 tubes. Each of which has a resolution of $130 \mu\text{m}$ [59].

3.2.4 Calorimeters

The ATLAS calorimeter system [60] consists of two sampling calorimeters, the electromagnetic calorimeter (ECal) and the hadronic calorimeter (HCal). The fine granularity ECal measures energies of electrons and photons with high precision. For jet reconstruction and E_T^{miss} measurements, the coarser HCal is employed.

Sampling calorimeters contain alternating passive absorption layers and active detection layers. The first is made from dense material, slowing the incoming particles and producing cascades of secondary particles. The energy of those particle showers is then measured in the active layers. The calorimeters are required to provide good containment for electromagnetic and hadronic showers, limiting the punch-through into the muon system.

Figure 3.7 presents the layout of the system, highlighting the different components of the ECal and HCal. As mentioned in section 3.2.2 a common vacuum vessel is used for solenoid and calorimeter system. The full azimuthal range is covered over a wide range in η , up to $|\eta| < 4.9$.

Electromagnetic calorimeter

Alternating layers of lead absorber material and liquid argon as active material are built in the ECal [62]. A highly-energetic electron or photon produces an electromagnetic shower in the lead plates. Those secondary particles ionize the argon gas, generating an electrical signal. The ECal features electrodes in a characteristic accordion structure,

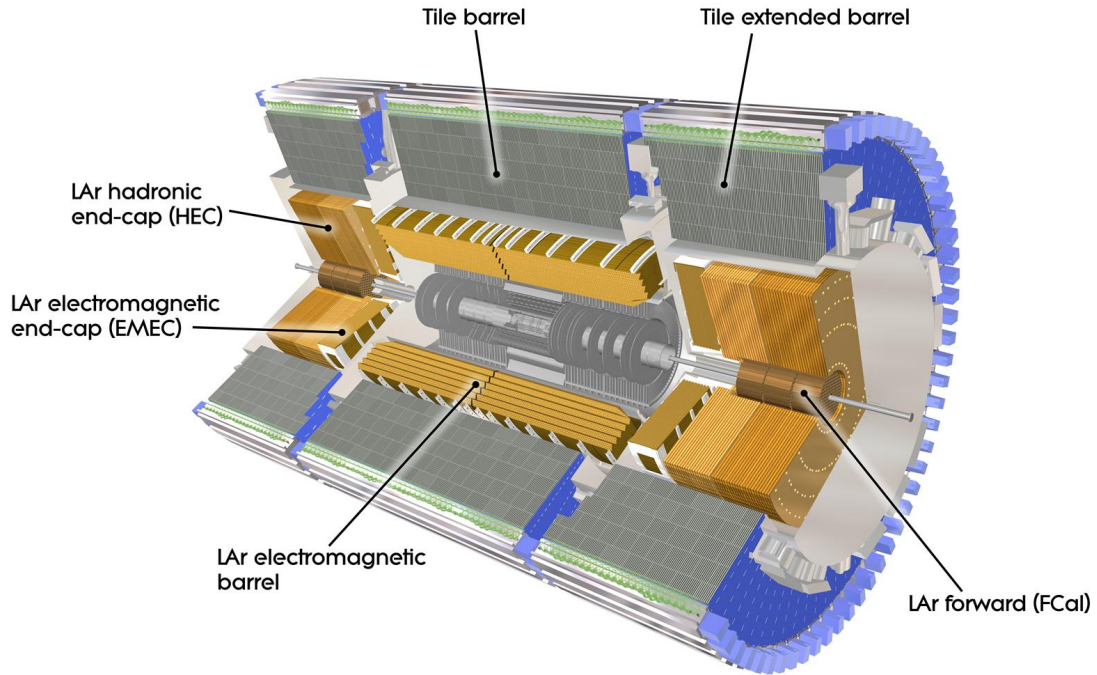


Figure 3.7: Schematic drawing of the ATLAS calorimetry system [61].

ensuring full azimuthal coverage. Constant gaps between the layers are ensured by varying the amplitude and folding angle of the accordion structures with the distance from the beam line.

The barrel calorimeter $|\eta| < 1.4$ consists of two identical, independent half-barrels. A 1 cm active liquid layer is added on the inner side of the barrel calorimeter. This presampler has the purpose to correct for energy lost in the coil and cryostat walls. The end-cap ECal consists of two concentric wheels, spanning the pseudorapidity interval $1.4 < |\eta| < 2.5$ and $2.5 < |\eta| < 3.2$, respectively.

Hadronic calorimeter

The HCal sits outside of the ECal to measure hadronic activity in an event which was not absorbed in the ECal. The barrel region of the HCal consists of the three-part Tile Calorimeter [63]. It employs steel absorber material and scintillating plates read out with wavelength shifting fibers. The fibers are oriented radially to collect the light from the tiles. A set of fibers is grouped into a photomultiplier to obtain a three dimensional segmentation of $\Delta\eta \times \Delta\phi = 0.1 \times 0.1$. The HCal has lower granularity compared to the ECal, which is compensated by its larger size. This ensures low probabilities that highly energetic particles pass through the calorimeter system undetected.

The tile calorimeter is designed as one barrel part ($|\eta| < 1.0$) and two extended barrels ($0.8 < |\eta| < 1.7$). A gap of 600 mm is needed between the parts to fit in infrastructure for the ID and LAr systems. To maximize the volume of active detector material, parts of that gap contain the Intermediate Tile Calorimeter. The angular range is extended for the region $1.7 < |\eta| < 3.2$ by the copper/liquid argon based hadronic endcap calorimeter (HEC) [62].

Forward calorimeter

The Forward Calorimeter (FCal) [62] extends the ATLAS calorimeter system in the forward direction $3.2 < |\eta| < 4.9$. It is installed in a particularly hostile region, due to the high level of radiation. The FCal also uses liquid argon as the active material and consists of three modules. Each module consists of a single absorber matrix carrying an array of tube electrodes in holes. Copper, sensitive to electromagnetic showers, is used as absorption material in the first module, The second and third module use tungsten, more suitable for hadronic showers.

3.2.5 Muon spectrometer

Muons are the only charged particles that do not lose a significant fraction of their energy when they traverse the detector. They penetrate ID and calorimeters to be detected in the outermost part of ATLAS, the Muon Spectrometer (MS) [64]. It is designed to provide precise measurements of muon tracks and momenta, independently of the ID. Similar to the ID, the momentum reconstruction is based on the magnetic deflection of muon tracks in a system of three large superconducting air-core toroid magnets.

The chambers are arranged in three cylindrical layers around the beam axis in the barrel region. The MS in the end-cap region is structured in three vertically installed layers (wheels). A computer generated image of the MS is given in Figure 3.8, while Figure 3.9 provides a more detailed illustration of the position of single components.

In addition to muon reconstruction, an essential requirement on the MS is the fast and efficient identification of muon candidates. These are used in the ATLAS hardware trigger which will be introduced in 3.2.6. To fulfill all requirements, the MS combines complementary tracking and triggering chambers.

Monitored Drift Tube Chambers

The Monitored Drift Tube (MDT) chambers [67] consist of 3–4 layers of pressurized drift tubes. Two multi-layers are mounted on either side of a support frame which also carries an optical monitoring system to correct for deformations.

The basic detection element of the MDTs are cylindrical aluminum drift tubes with 30 mm outer radius and 400 μm wall thickness. A sense wire is positioned in the center of each tube with an accuracy of 20 μm . The MDTs operate with a Ar/CO₂ (93%/7%) gas mixture at a pressure of three bar. A muon passing a tube ionizes the

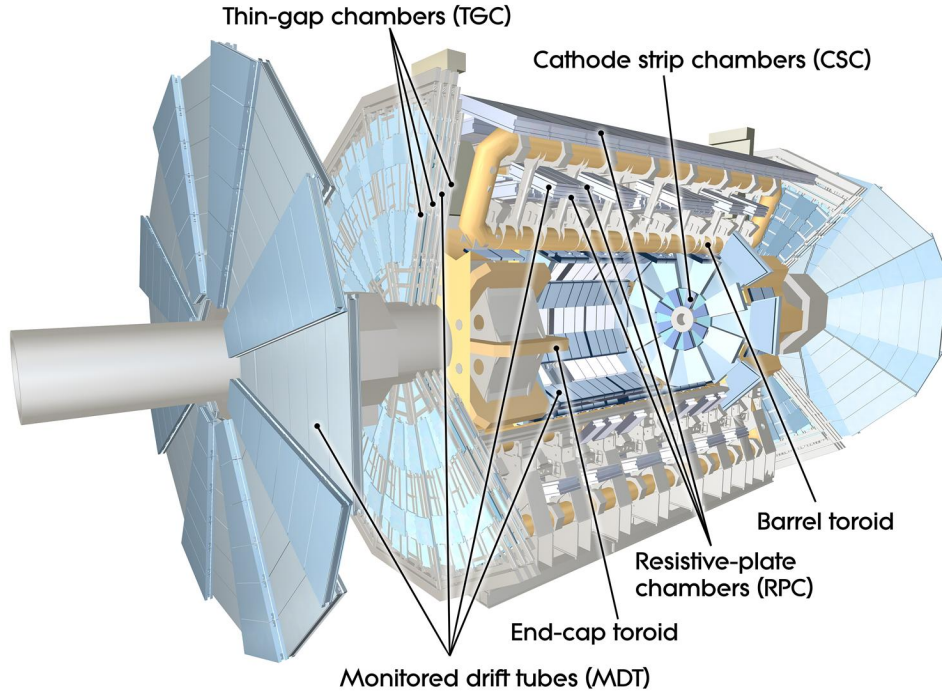


Figure 3.8: Configuration of the muon spectrometer with its four chamber sub-systems [65].

gas mixture. The resulting electrons start drifting towards the anode wire producing secondary electrons in the process. Measuring the drift time of the ionized electrons allows the reconstruction of the position of the muon hit perpendicular to the tube.

The MDT chambers in the barrel region ($|\eta| < 1.3$) are arranged in 16 sectors. To achieve full coverage in φ , the chambers are slightly overlapping. The drift tubes are oriented parallel to the beam axis, ensuring excellent momentum resolution in the transverse plane. MDT chambers are also installed in the MS wheels in the end-cap region, covering up to $|\eta| < 2.7$.

Cathode Strip Chambers

While the MDT chambers are well suited for the precise muon reconstruction requirements of the MS, they are not suited for areas where high counting rates ($> 200 \text{ Hz/cm}^2$) are expected. Such high background rates are expected in the first wheel at $|\eta| > 2.0$ [64]. In this region of the spectrometer, an alternative detector concept is employed, the Cathode Strip Chamber (CSC) [68].

The CSCs are multi-wire proportional chambers with segmented cathode readout. They combine excellent spatial resolution with high counting rate capabilities. The CSCs have a symmetric cell design in which the spacing between the cathode strips and the

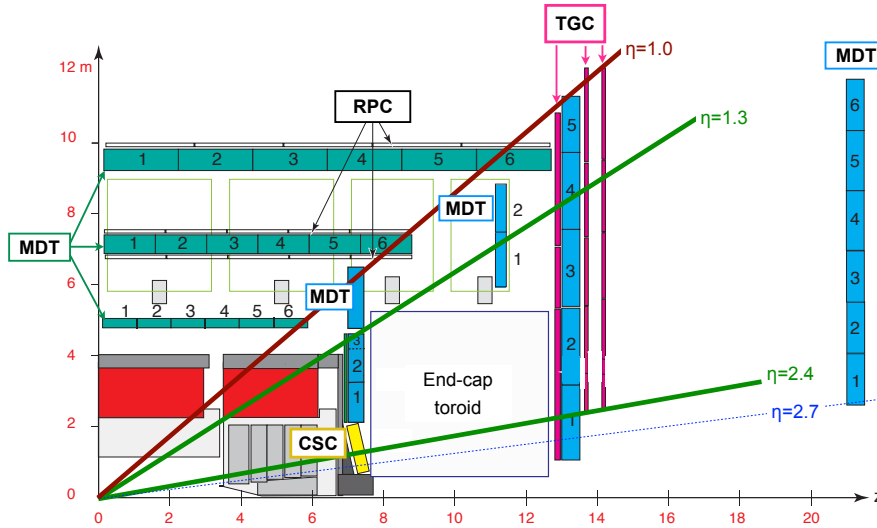


Figure 3.9: Cross-section of the muon system in a plane containing the beam axis. Infinite-momentum muons would propagate along straight trajectories with fixed η . Figure taken from [66].

anode-wires is equal to the distance between wires. They are operated with Ar/CO₂ gas mixture (80%/20%).

One of the cathodes is segmented with the strips perpendicular to the wires, while the strips on the other cathode are oriented parallel. 16 four-layered CSCs are installed in the first wheel in each end-cap.

Resistive Plate Chambers

Apart from precise muon reconstruction, the main task of the MS is to provide input to the trigger. A system of Resistive Plate Chambers (RPCs) [69] is used in the barrel region (up to $|\eta| < 1.05$) for that purpose. They are designed to be fast and hence have low-occupancy.

The RPC is a gaseous detector consisting of two parallel resistive plates and insulating spacers defining the 2 mm gas gap. The gas mixture is C₂H₂F₄/C₄H₁₀/SF₆ (94.7%/5%/0.3%). The discharge electrons, produced by a high-energy muon, drift in the gas towards the plates. A homogeneous electrical field is applied, amplifying the signal which is finally collected by copper strips.

The RPCs are assembled together with a MDT of equal dimensions in a common

mechanical support structure. Each RPC consists of two independent detector layers. Two orthogonal sets of read-out strips in each chamber measure the longitudinal (η) and transverse (φ) coordinates. They contribute to the track reconstruction by measuring the coordinate along the MDT tubes in the non-bending plane.

Thin Gap Chambers

The final component of the MS are the Thin Gap Chambers (TGCs). Their purpose in the end-cap region is two-fold: provide muon trigger capabilities as well as the azimuthal coordinate to complement the MDT measurements. The TGCs have a structure similar to CSCs. However, the distance between the anode-wires is larger than the distance between wire and cathode plate. The TGCs are designed to have excellent time resolution which allows to assign more than 99% of the muons to the correct bunch crossing [64].

3.2.6 Trigger and data acquisition

The Trigger and Data Acquisition (TDAQ) system [70] is an essential component of the ATLAS experiment. A decision whether data from a given collision is recorded for offline analysis has to be made in real time by the system. Events that are discarded in the trigger chain are lost forever. Thus, it is crucial that the ATLAS TDAQ system must be efficient and robust to fully exploit the physics potential of the LHC.

At a bunch crossing frequency of 40 MHz, storing the collision data of every event would amount to more than 50 TB of data per second. Reading out and storing this volume of data is computationally not feasible. In addition to the immense scale of the data set, a large fraction of these events do not contain the specific physics signatures which are of interest for further investigation. ATLAS employs a two-level trigger system to reduce the average rate to record collision data to 1 kHz. This brings the data storage rate of the detector to manageable levels. The operation of the trigger system during the challenging Run-2 conditions is summarized in [71]. A schematic diagram highlighting the relevant components for triggering as well as the detector read-out and data flow is shown in Figure 3.10.

Level-1 Trigger

The Level-1 (L1) trigger searches for events with high transverse momentum particles. It selects events by considering event-level quantities, such as the total energy in the calorimeter, the multiplicity of objects above a certain threshold or invariant masses and angular distances between objects.

The selection is based on information retrieved from a subset of detectors. The L1 muon (L1Muon) trigger uses signals from RPCs and TGCs. Coincidence requirements between outer and inner TGC stations are applied to reduce the rate of particles not originating from the primary interaction point. Calorimeter information on the other hand enters the L1 calorimeter (L1Calo) trigger. Input from both subsystems is passed

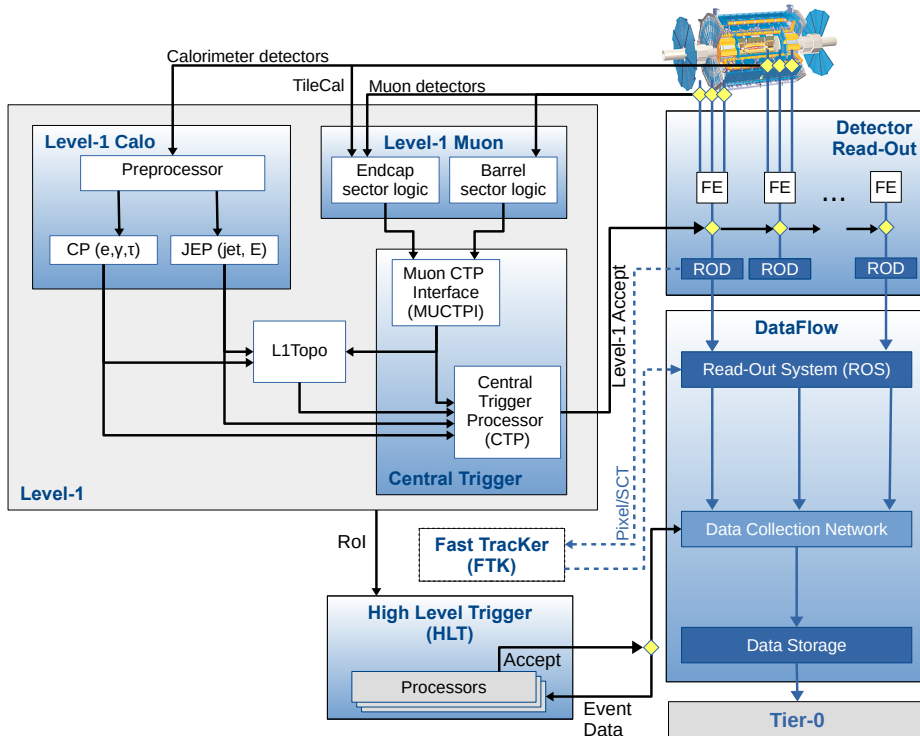


Figure 3.10: The ATLAS TDAQ system as it was used in Run 2. Components relevant for triggering as well as the detector read-out and data flow are shown. Figure taken from [71].

to the Central Trigger Processor which forms the final L1 trigger decision.

Furthermore, the L1 trigger identifies Regions of Interest (RoI's) in η and ϕ within the detector. These will be investigated further by the second trigger stage. The L1 trigger accepts events at a rate up to the maximum detector read-out rate of 100 kHz [71].

High Level Trigger

The High Level Trigger (HLT) is the software-based second stage of the trigger. It uses all the available detector data at full precision and granularity within the RoI's identified by the L1 (approximately 2% of the total event data) [38].

Dedicated fast trigger algorithms providing early rejection criteria are followed by more precise (and CPU intense) algorithms. They explicitly reconstruct physics objects and apply kinematic selections to them. They are processed on a dedicated computing farm.

Events are finally selected by *trigger chains* consisting of a L1 trigger item and a series of HLT algorithms. Each chain is designed to select a particular physics signature such

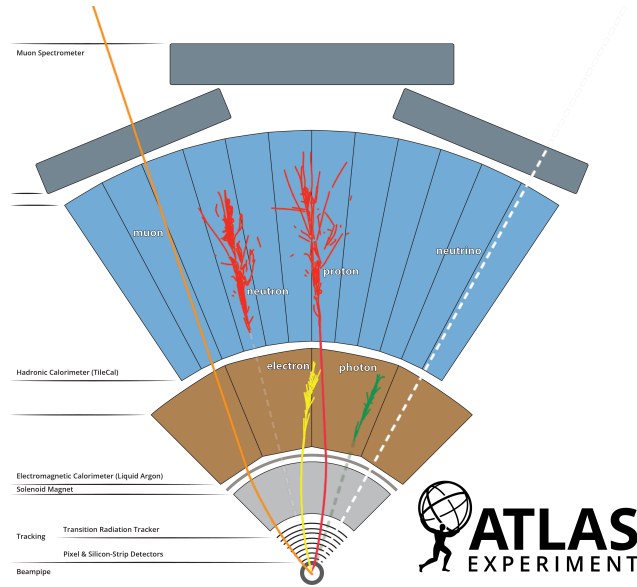


Figure 3.11: Schematic diagram of the ATLAS detector illustrating the signal that various particles induce in the detector. Dashed lines indicate no measurable signal [73].

as the presence of leptons, photons, jets, missing transverse momentum or total energy in an event.

Once an event is accepted by the HLT, the data is permanently stored for offline reconstruction and exported to the Tier-0 facility [72] at CERN's computing center. The physics output rate of the HLT is on average 1.2 kHz corresponding to an average physics throughput to permanent storage of 1.2 GB/s [71].

3.2.7 A particle's journey through ATLAS

To summarize this chapter on the ATLAS experiment, an illustrative cutview of the detector is shown in Figure 3.11. The traces that several particle species leave in sub-components of the machine are

- **Electrons** leave curved tracks in the ID due to the magnetic field. They produce an electromagnetic shower in the ECal, depositing most of their energy there.
- **Photons** do not carry electrical charge and are therefore not detected in the ID. Like electrons, they are absorbed in the ECal.
- **Muons** tracks can also be reconstructed in the ID. However, they do not lose a significant amount of energy in the calorimeters. The dedicated MS provides precise energy and location information.

- **Hadrons** may leave tracks in the ID if they are charged. They primarily interact strongly in the HCal and are reconstructed as jets.
- **Neutrinos** have a negligible cross section with the detector material and leave no direct signal. Their presence is inferred from missing transverse energy in the event.

4 Object Reconstruction

The raw data output of the ATLAS experiment consists of electrical signals from ≈ 40 million channels. Particle trajectory and energy cluster information are combined in a Particle Flow (PFlow) algorithm to construct objects that serve as input to jet clustering algorithms. Objects, such as jets, leptons and E_T^{miss} are reconstructed as high-level objects for physics analyses. The objects are subsequently calibrated to account for differences in MC simulation and data and to ensure accurate and precise measurements of their properties.

This chapter details the reconstruction of tracks, calorimeter cluster and physics objects within the ATLAS detector. The subsequent calibration will be discussed for the example of b tag calibration in Chapter ... The calibration of other objects will not be discussed in detail.

along with the reconstruction and calibration procedures for physics objects used in analysis.

4.1 Reconstruction of Tracks, Clusters and Vertices

This section provides a concise summary of low level objects that will serve as an input for the reconstruction of physics objects.

Tracks

As illustrated in Figure 3.11, charged particles traversing the various layers of the ATLAS detector produce spatial hits and energy depositions. The particles are forced on helical trajectories by the strong magnetic field, enabling the determination of momentum and charge. The trajectories are reconstructed from ID and MS signals to form tracks.

The first step in the reconstruction of charged particles is the creation of three-dimensional *space points* from neighboring cells in Pixel and SCT that exceed a certain threshold [74].

Sets of three space points form *track seeds* which allow a first crude momentum estimate while maximizing the number of possible combinations. The candidates are extrapolated by assuming a perfect helical trajectory in a uniform magnetic field to estimate the distance to beam line. These are also called *transverse impact parameter* d_0 and *longitudinal impact parameter* z_0 .

After applying a first set of momentum and impact parameter requirements, a combinatorial Kalman filter [75] is used to build track candidates from the chosen seeds in an

iterative fitting process. It incorporates additional space-points from the remaining layers of the Pixel and SCT detectors which are compatible with the preliminary trajectory. Track candidates created from random collection of space points are rejected to manage the available CPU budget for event reconstruction.

A *track score* is defined evaluating tracks according to their momentum, number of clusters and χ^2 of the fit, where more energetic tracks with an increased number of clusters are ranked higher. The tracks are processed by an ambiguity solver in descending order of the track score to resolve overlapping space points.

The ambiguity solver rejects all tracks that fail a number of criteria including kinematic cuts ($p_T > 400$ MeV, $|\eta| < 2.5$) and on the total number of clusters and holes. A hole is defined as an intersection of the reconstructed track trajectory with a sensitive detector element that does not contain a matching cluster. The detailed list can be found in [74].

The tracks can be extended in the outer TRT in the next step. An extension algorithm searches for seeded candidates before a second module processes and evaluates these extensions. It should be noted, that this is a pure extension, not a combination. The silicon-only tracks are not modified [76].

The first step in the reconstruction of muons in the MS [77] is to form segments from hit patterns in the muon chambers. A Hough transform [78] is used to search for aligned hits in MDT and trigger chambers. It speeds up the reconstruction and makes it more robust against misidentification.

A straight line is fitted to the hits in each layer is performed to reconstruct the MDT segments. Muon track candidates are then built by combining hits from segments in different layers. The segment-seeded combinatorial search starts from the middle layers of the detector where more trigger hits are available, and is then extended to use the segments from the outer and inner layers. Finally, a global χ^2 fit is performed to the hits associated with each track candidate.

Calorimeter cluster

The reconstruction of particles depositing energy in the calorimeter system is based on a three-dimensional topological clustering of individual calorimeter cell signals [79]. The resulting topological cell clusters (*topo-clusters*) carry both, shape and location information.

The cell significance $\zeta_{\text{cell}}^{\text{EM}}$ is defined to control the cluster formation

$$\zeta_{\text{cell}}^{\text{EM}} = \frac{E_{\text{cell}}^{\text{EM}}}{\sigma_{\text{noise,cell}}^{\text{EM}}}, \quad (4.1)$$

where $E_{\text{cell}}^{\text{EM}}$ and $\sigma_{\text{noise,cell}}^{\text{EM}}$ denote the cell signal and average (expected) noise, respectively. The algorithm to form *topo-clusters* starts from seeds with highly significant signal $|\zeta_{\text{cell}}^{\text{EM}}| > 4$ (primary seed threshold). Each of the seed cells forms a *proto cluster*. All neighboring cells with $|\zeta_{\text{cell}}^{\text{EM}}| > 2$ (growth threshold) are collected iteratively, as well as

cells with $|\varsigma_{\text{cell}}^{\text{EM}}| > 0$ (principal cell filter). In this case, neighboring cells refers to cells adjacent in a given sampling layer, as well as cells with partial overlap in adjacent layers.

If a proto cluster has more than one local maximum it is split into separate clusters. A local maximum is defined as a cell with $E_{\text{cell}}^{\text{EM}} > 500 \text{ MeV}$ and at least four neighbors, none of them with larger signal.

The resulting proto-cluster is characterized by a core of seed cells, surrounded by an envelope of cells with less significant signals. The energy of the cluster is given by the energy sum of the associated cells taking geometrical signal weights into account, while η and φ are calculated from the energy-weighted average of the cells. The clusters are assumed to be massless pseudo-particles.

Vertices

As mentioned in Section ... , there are multiple proton-proton collisions per bunch crossing. In order to identify the one collision of interest, the intersection of multiple tracks, referred to as *vertex* is reconstructed [80]. The vertices at the beam line are denoted as *primary vertices*, where the one with the largest sum of squared transverse momenta of associated tracks is labeled the *hard-scatter vertex*.

Secondary vertices are displaced with respect to the beam line. They can be an indication for long-lived particles such as *b* hadrons, that travel a short distance in the detector before decaying.

4.2 Lepton reconstruction

Leptons are reconstructed using either tracks or a combination of tracks and topo-clusters. This discussion focuses on the reconstruction and calibration of electrons and muons, as tau leptons are not considered in the work presented in Part II or the analysis in Part III. However, the results may include events where electrons and muons originate from tau lepton decays.

4.2.1 Electron Reconstruction

Offline electron reconstruction starts from dynamic, variable size clusters in the ECal. The seed clusters need to exceed $E_T > 1 \text{ GeV}$ and be matched to a track with at least four space points [81]. Clusters within $\Delta\eta \times \Delta\varphi = 0.125 \times 0.300$ that are matched to the same track tend to represent secondary EM showers originating from the same initial electron and are called *satellite clusters*. Seed clusters with their associated satellite clusters constitute *superclusters*.

In order to distinguish between *prompt* electrons, that can originate from a *W* or *Z* boson decay from the primary vertex, *non-prompt* electrons that can be produced in hadron decays, photons and low energetic jets, a set of discriminating observables is defined [81, 82]. These observables are defined in the region covered by the ECal

($|\eta| < 2.47$) and include the number of hits in the ID subsystems and the ration of energies in the calorimeter system. They can be used to define the three working points (WPs) Loose, Medium and Tight that impose increasingly strict requirements. Therefore, the Tight WP exhibits the strongest background suppression, at the cost of a loss in efficiency.

To suppress electrons originating from heavy hadron decays, an isolation criteria is applied, checking for detector activity in a cone ($\Delta R < 0.2$) around the electron. The PLImprovedTight isolation working point [83] is employed.

It is vital to account for performance differences in identification, reconstruction and isolation between data and simulation. By comparing the respective efficiencies in $Z \rightarrow e^+e^-$ and $J/\psi \rightarrow e^+e^-$ events scale factors (SFs) can be obtained [81]. The SFs are applied to the MC weights to better describe the data.

The absolute energy scale and the energy resolution is corrected using $Z \rightarrow e^+e^-$ events. Validation of the calibration is performed with $J/\psi \rightarrow e^+e^-$ and radiative Z processes. The uncertainties assigned to the calibration are typically 0.05% for electrons from resonant Z decays [84].

Requirement	Signal electrons (tight)	Background electrons (loose)
Identification	LHTight	LHLoose
Isolation	PLImprovedTight	fail PLImprovedTight or fail tight selection
p_T cut	$p_T > 10 \text{ GeV}$	$p_T > 10 \text{ GeV}$
η cut	$ \eta < 2.47$	$ \eta < 2.47$
Crack veto	veto $1.37 < \eta < 1.52$	veto $1.37 < \eta < 1.52$
$ d_0 /\sigma_{d_0}$ cut	$ d_0 /\sigma_{d_0} < 5.0$	$ d_0 /\sigma_{d_0} < 5.0$
$ z_0 \sin(\theta) $ cut	$ z_0 \sin(\theta) < 0.5 \text{ mm}$	$ z_0 \sin(\theta) < 0.5 \text{ mm}$
Bad cluster veto	yes	yes

Table 4.1: A summary of the electron definitions used in the analysis. Signal electrons are used for the cross section measurement, while background electrons are used in the overlap removal. d_0 and z_0 are the transverse and longitudinal impact parameters introduced in Section 4.1

4.2.2 Muon Reconstruction

Muons are minimum-ionizing particles. Their signature in the detector includes tracks in ID and MS. The characteristic energy deposits in the calorimeter are also used to account for cases with large energy loss in the calorimeter system and MS-independent identification. The information provided by ID, MS and calorimeters is combined, defining five categories of muons [85]:

- Combined (CB) muons are formed with a global refit that uses the independent reconstructions in ID and MS.

- Inside-out combined (IO) muons are reconstructed using a complementary algorithm that extrapolates ID tracks to the MS. Reconstruction can be improved in regions with limited MS coverage, as well as for low- p_T muons, that may be stopped in the MS.
- Segment-tagged (ST) muons classify an ID track as a muon if it is associated with at least one local track segment in the MDT or CSC.
- Calorimeter-tagged (CT) muons match a track in the ID with an energy deposit in the calorimeter compatible with a minimum-ionizing particle. The muon p_T threshold has to be increased from 2 GeV to 5 GeV due to large background contamination at low p_T .
- Extrapolated (ME) muons are reconstructed from MS hits only, adding a loose requirement that the origin of the track is compatible with the IP. Thus, extending the coverage to $|\eta| < 2.7$ to exploit the full MS range.

Overlaps between the categories are resolved before producing the collection of muons used in physics analyses. Similar to the electron identification, a *Loose*, *Medium* and *Tight* WP is defined for muons to suit the needs of the wide variety of physics analyses involving final states containing muons. The *Medium* WP, which will be used in the analysis presented in Chapter ... only accepts CB and IO muons within the ID acceptance $|\eta| < 2.5$. The fraction of CB muons passing the *Medium* WP in $t\bar{t}$ events exceeds 98% [85].

Again, a calibration procedure is defined and applied to simulated data to match the performance measured in real data [86]. Samples of $J/\psi \rightarrow \mu^+\mu^-$ and $Z \rightarrow \mu^+\mu^-$ events are used to measure the momentum scale and resolution. The momentum scale and resolution are measured with an uncertainty of 0.5% and 1.5% at the Z peak. Charge dependent biases are removed with a dedicated correction.

Systematic uncertainties are defined for variations in the SFs associated with decay and final state radiation modeling, J/ψ only and Z only fit or statistical uncertainties. The full list is given in [86].

4.3 Jet Reconstruction

Quarks and gluons cannot be reconstructed as single particles in the detector. Due to QCD color confinement, they hadronize into collimated sprays of color-neutral particles, which are reconstructed as jets.

Particle flow

ATLAS constructs jets from Particle Flow (PFlow) objects [87], which optimally combine tracking and calorimeter information. Integrating tracking significantly improves the energy and angular resolution for low-momentum particles, below typical calorimeter

Requirement	Signal muons (tight)	Background muons (loose)
Quality	Medium	Medium
Bad muon veto	yes	yes
Isolation	PLImprovedTight	fail PLImprovedTight
p_T cut	$p_T > 10 \text{ GeV}$	$p_T > 10 \text{ GeV}$
η cut	$ \eta < 2.5$	$ \eta < 2.5$
$ d_0 /\sigma_{d_0}$ cut	$ d_0 /\sigma_{d_0} < 3.0$	$ d_0 /\sigma_{d_0} < 3.0$
$ z_0 \sin(\theta) $ cut	$ z_0 \sin(\theta) < 0.5 \text{ mm}$	$ z_0 \sin(\theta) < 0.5 \text{ mm}$

Table 4.2: Summary of the muon definitions in the analysis. Signal muons are used for the cross section measurement, while background muons are used in the overlap removal. d_0 and z_0 are the transverse and longitudinal impact parameters introduced in Section 4.1

noise thresholds. Furthermore, track-based vertex association allows effective rejection of pile-up contributions.

The PFlow algorithm begins by selecting well-measured tracks and matching them to calorimeter topo-clusters. Overlaps between the momentum and energy measurements made in the inner detector and calorimeters are removed by a cell-based energy subtraction algorithm. The expected calorimeter energy deposit from the particle that also created the track is estimated from the track momentum and and topo-cluster position.

Since single particles often deposit energy across multiple topo-clusters, the algorithm evaluates shower overlap probabilities and adds relevant clusters where necessary to recover the full shower energy. The expected track energy is then subtracted cell-by-cell from the matched topo-clusters. Any remaining calorimeter energy inconsistent with single-particle shower fluctuations is discarded, yielding a clean list of PFlow objects.

Jet clustering

The PFlow objects are subsequently clustered into jets using the anti- k_T algorithm [88]. A sequential recombination algorithm is defined introducing distances d_{ij} between entities (particles, pseudojets) i and j and d_{iB} between entity i and the beam (B)

$$d_{ij} = \min(k_{ti}^{2p}, k_{tj}^{2p}) \frac{\Delta_{ij}^2}{R^2}, \quad (4.2a)$$

$$d_{iB} = k_{ti}^{2p} \quad (4.2b)$$

where $\Delta_{ij}^2 = (y_i - y_j)^2 + (\varphi_i - \varphi_j)^2$ and k_{ti} , y_i and φ_i are respectively the transverse momentum, rapidity and azimuth of particle i . The jet radius parameter R controls the jet size.

The inclusive sequential recombination algorithm iteratively selects the smallest distance from a set of pairwise (d_{ij}) and beam (d_{iB}) measures. If the minimum is

between entities (d_{ij}), they are merged into a single object, otherwise (d_{iB}), the entity is declared a final jet and removed from the list. The distances are updated and the procedure repeats until all particles are clustered into jets.

The choice of p defines jet algorithm, where $p = 1$ corresponds to the k_t [89], $p = 0$ to the Cambridge/Aachen [90] and $p = -1$ to the anti- k_t algorithm. The latter is generally used in ATLAS analyses, as it constitutes a simple, IRC safe and soft-resilient jet algorithm, constructing jets whose shape is not influenced by soft radiation. The hardest particles act as seeds for the reconstruction reducing sensitivity to pileup and UE effects [88]. The analysis presented in Chapter ... employs jets clustered from PFlow objects with the anti- k_t algorithm and $R = 0.4$. The majority of pile-up jets are rejected using the jet vertex tagger (JVT) tool [91].

b tagging

In order to measure cross sections with the $WbWb$ final state, it is vital to identify jets originating from b quarks. b jet identification in ATLAS exploits the distinct decay topology of b hadrons. Their longer lifetimes and larger masses compared to light-flavor or charm hadrons, result in measurable displaced decay vertices. The algorithm relies on ID track information, primarily transverse impact parameters and reconstructed secondary or tertiary vertices. The DL1r [92] algorithm combines multiple low-level taggers into a single discriminant using a deep neural network. Further improvements are obtained by exploiting a recurrent neural network in the RNNIP algorithm [93], significantly enhancing discrimination against light-flavor and c jet backgrounds.

Several WPs are defined to balance b jet tagging efficiency against c jet and light flavor rejection. For this analysis, the DL1r tagger at WP 77% is employed. The performance of the low level taggers compared to the combined DL1r tagger is highlighted in Figure 4.1.

ATLAS flavor taggers are trained on simulated MC samples. The calibration procedure is detailed in Chapter ...

Requirement	Signal jets	Baseline jets
Jet type	AntiKt4EMPFLOWJETS	AntiKt4EMPFLOWJETS
JVT working point	Tight	Tight
p_T cut	$p_T > 25 \text{ GeV}$	$p_T > 20 \text{ GeV}$
η cut	$ \eta < 2.5$	$ \eta < 4.5$
b -tagging	DL1r with FixedCutBEff_77	

Table 4.3: Summary of the jet definitions in the analysis.

Jet energy scale

A multi-step calibration framework is implemented to derive corrections that account

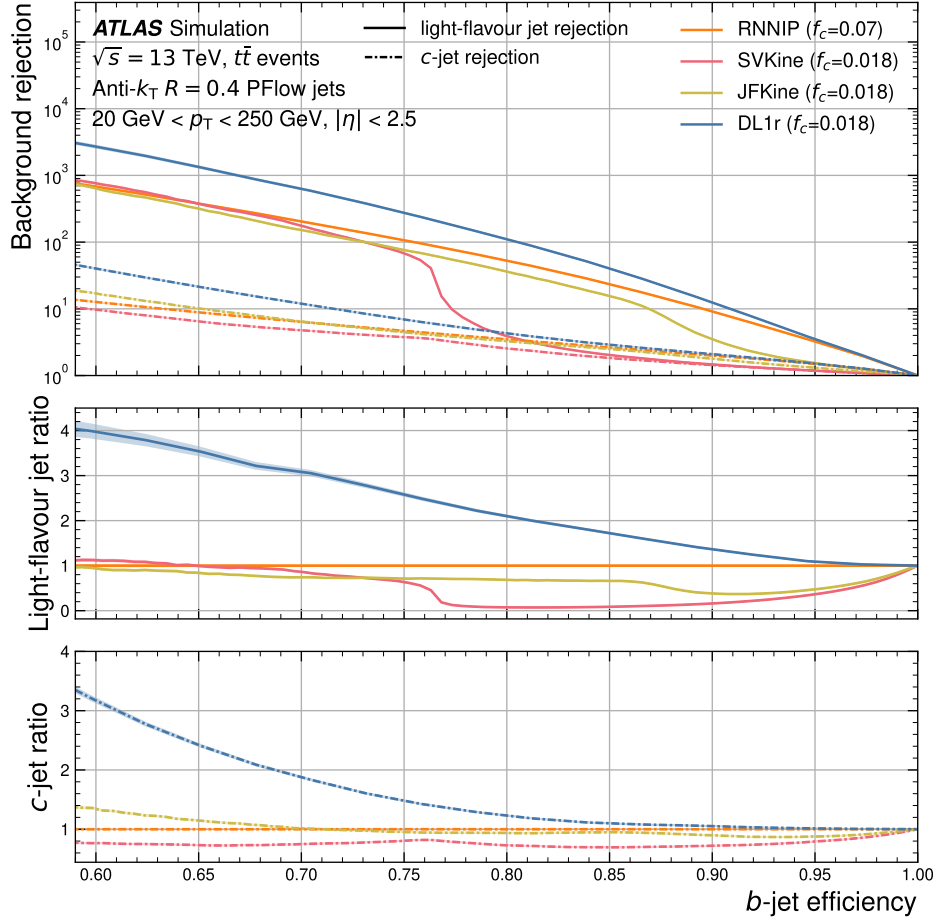


Figure 4.1: The light flavor jet and c jet rejection factors as a function of the b tagging efficiency for DL1r as well as the low-level taggers RNNIP, SVKine, and JFKine. The ratio of light and c jet rejection rates are shown in the two lower panels. f_c denotes the effective c jet fraction in the background hypothesis [92].

for detector response, reconstruction biases, and environmental effects in the *Jet Energy Scale* (JES) [94]. The complete calibration chain is summarized in Figure 4.2. Throughout all calibration stages, the jet four-momentum is consistently corrected by scaling jet p_T , energy, and mass.

To remove excess energy from additional pp interactions (pile-up) from an event, a correction based on p_T density and jet area is derived from MC simulation. The second step is a residual correction parameterized as a function of μ and the number of reconstructed primary vertices in the event (N_{PV}).

The absolute JES calibration corrects the 4-momentum of the jets in bins of p_T and η to account for energy losses in passive material and biases in the reconstruction. Reconstructed jets are geometrically matched to truth jets to derive the calibration from

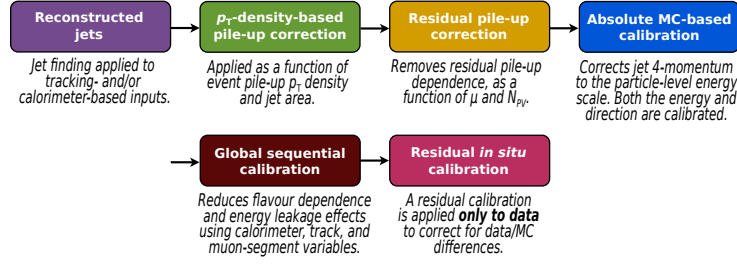


Figure 4.2: Illustration of the several jet energy calibration steps. Each correction is applied to the jet four-momentum [94].

simulations of dijet events.

The global sequential calibration removes the dependence on observables constructed using information from the tracking, calorimeter, and muon systems. The discrepancies originate from the different average energy distributions and constituent multiplicities of quark or gluon-initiated jets. It improves the jet resolution without changing the average jet energy response.

All these calibrations are applied to both data and MC simulation.

To correct for remaining differences between data and MC simulation a three staged in-situ calibration is applied to the data only. The differences are caused by an imperfect simulation of both detector materials and underlying physics processes, including pile-up, jet formation or interaction of particles with the detector material.

The first step corrects the energy scale of forward jets ($0.8 < |\eta_{\text{det}}| < 4.5$) to match those of central jets ($|\eta_{\text{det}}| < 0.8$). Subsequently, the hadronic recoil in an event is balanced against well calibrated Z bosons or photons. The same concept is applied in the last step, where a system of well-calibrated low- p_T jets is used to calibrate single high- p_T jets.

An extensive list of systematic uncertainties with 125 components is defined for the JES [94] including uncertainties on the pile-up and flavor composition modeling, as well as dependence on the reference objects used in the calibration. Several sources can be combined, taking the respective correlations between the JES uncertainties of two jets at different p_T and η into account. The CategoryReduction scheme [95] is used in the measurement. It provides roughly 30 independent variations that parametrize the uncertainties in the JES.

The JES systematic uncertainty is one of the dominating uncertainties for jet based observables. For central jets it ranges from 1% up to 5% close to the p_T threshold at 20 GeV [94].

Jet energy resolution

In addition to a precise understanding of the JES and the associated uncertainties, the *Jet*

Energy Resolution (JER). In order to measure the JER, two jets need to be balanced against each other in a well-defined dijet system. It can be parametrized using a functional form with three independent terms related to noise (N), stochastic effects (S) and a constant offset (C) [94]

$$\frac{\sigma(p_T)}{p_T} = \frac{N}{p_T} \oplus \frac{S}{\sqrt{p_T}} \oplus C \quad (4.3)$$

The noise term accounts for fluctuations from electronics and pile-up. It is independent of the energy deposited in the calorimeters and is expected to be significant for low energetic jets. Direct estimations can be obtained with the *random cone* method described in Reference [94]. The stochastic term scales with $1/\sqrt{p_T}$ and accounts for fluctuations in the amount of energy deposited. It represents the limiting term in the resolution. The dominating term at high momenta ($p_T > 400$ GeV) is a constant contribution corresponding to fluctuations in the starting point of the hadron shower as well as a non-uniform response across the calorimeter and energy depositions in the passive material.

A smearing procedure is introduced to ensure that the resolution of the jet energy scale in simulation matches that in data [94]. The uncertainty in the jet energy resolution is assessed using the FullJER scheme [95] providing 13 independent variations.

4.4 MET Reconstruction

In an ideal collision, the protons collide head-on along the beam line with a vanishing momentum component in the transverse plane. An imbalance in the final state transverse momentum of an event therefore indicates particles that do not leave a measurable signals in the detector. The missing transverse energy E_T^{miss} is typically associated with neutrinos in physics analyses that do not consider BSM processes.

Two contributions characterize the reconstructed E_T^{miss} [96]. The fully reconstructed and calibrated leptons, photons and jets enter, being part of the *hard event*. The second contribution comes from *soft event* signals consisting of calibrated tracks associated with the hard scatter vertex, that are not associated to any other reconstructed objects. E_T^{miss} is constructed from the transverse momentum vectors of the various contribution as the negative vector sum.

Systematic uncertainties on E_T^{miss} are driven by the energy scale and resolution of all objects contributing to the soft and hard terms. The hard term composition is analysis-specific and uncertainties are extracted per object type using official ATLAS recommendations. Uncertainties for identical object types are treated as fully correlated, while those for different object types are assumed uncorrelated, reflecting the independent reconstruction and calibration chains.

4.5 Overlap removal

To mitigate double counting of energy deposits arising from objects that are reconstructed twice (e.g., an electron also forming a jet), ambiguous objects are resolved using an overlap removal procedure [97]. Loose object definitions listed in Tables ?? are applied for electrons, muons, and jets for this step. The overlap removal algorithm resolves overlaps in the following order:

- Remove PFlow jets that are actually muons.
- Muons overlapping with electrons that leaves a sufficiently high energy deposit in the calorimeter and share the same ID track. Any electron sharing a track with the remaining muons is removed.
- Jets within $\Delta\eta < 0.2$ of an electron are removed. Electrons within $\Delta\eta < 0.4$ of the remaining jets are discarded.
- muon-jet: Jets within $\Delta\eta < 0.2$ of a muon are removed if they feature less than three tracks or fulfill $p_{T,\mu}/p_{T,\text{jet}} > 0.5$ and $p_{T,\mu}/\Sigma p_{T,\text{tracks}} > 0.7$ are removed. Muons overlapping with $\Delta\eta < 0.4$ of the remaining jets are removed.

4.6 Definition of physics objects on particle level

The aim of this section is to provide definitions of physics objects, such as leptons, jets and missing momentum, that allow for a direct comparison of theoretical and experimental results. The ATLAS collaboration provides recommendations [98] for particle level definitions with a minimal model dependence. They should be accurate, unambiguous and allow the comparison of predictions with measurement results between experiments without extensive prior experimental knowledge. The specified particle level objects are as close to experimentally reconstructed objects as possible, avoiding large extrapolations to regions outside of the detector acceptance.

- All **stable particles**, defined by a lifetime $c\tau_0 > 10$ mm within the angular acceptance $|\eta| < 5$ are considered.
- **Prompt leptons** are generated final-state neutrinos, electrons and muons not originating from hadron decays. Charged leptons from subsequent τ decays ($W \rightarrow \tau\nu_\tau \rightarrow \ell\nu_\ell\nu_\tau$) are therefore also categorized as prompt. To take QED FSR into account, the charged leptons are *dressed* by clustering all photons within a cone around the *bare* lepton (after simulating QED FSR) with the anti- k_t algorithm. The clustering distance parameter is set to $R = 0.1$.
- **Jets** are built in a subsequent step by collecting all stable particles apart from the identified prompt dressed leptons and prompt photons. The anti- k_t algorithm with $R = 0.4$ is employed for the jet building. The jet is labeled as a b jet if a b

hadron with $p_T > 5$ GeV is found within $\Delta R < 0.3$ around the jet axis. Otherwise it is labeled as a *light* jet. No jet-lepton overlap removal is applied.

- **Missing transverse momentum** E_T^{miss} corresponds to the transverse vector sum of all neutrinos not from hadron decays.

5 Calibration of ATLAS flavor taggers

Describe the QT including some plots on the calibration.

6 Measurement of $WbWb$ cross sections

Introduction to the measurement.

6.1 Baseline Event selection

The phase space regions of interest, also known as signal regions, are defined by the complex final state topology of the $WbWb$ process. The signal consists of all processes resulting in a $WbWb$ final state. However, contributions from diboson production in association with a b quark are considered negligible. Exemplary diagrams of the two dominant process, $t\bar{t}$ and tW production are depicted in Figure ...

The semileptonic final state is considered in this thesis, where one W boson decays leptonically, while the other decays hadronically. The $WbWb$ final state is therefore defined by

$$pp \rightarrow WbWb \rightarrow \ell \nu q \bar{q}' b \bar{b}$$

where the lepton is either an electron or a muon. In order to enhance the signal process and reduce background contamination, a variety of initial requirements (cuts) are applied. The event selection is optimized using simulated MC event samples.

The same cuts are applied on particle and detector level. This way extrapolating into phase spaces that cannot be measured is avoided.

- Exactly one signal lepton, as defined in ... is required. Events are selected with a single lepton trigger. It has to exceed $p_T > 30$ GeV and fulfill $|\eta| < 2.5$.
- Since the neutrino cannot be detected, significant E_T^{miss} is expected in the event. It is combined with the transverse mass of leptonically decaying W boson m_T^W computed from the lepton and E_T^{miss} 4-vectors where the z-component is set to 0. The final cut at $E_T^{miss} + m_T^W > 60$ GeV aims to suppress QCD background, with a mid-identified or non-prompt lepton.
- At least three jets reconstructed with the anti- k_t clustering and a jet radius parameter of $R = 0.4$, as defined in Chapter ... , are required. They need to pass $p_T > 25$ GeV and $|\eta| < 2.5$. At least two of them have to fulfill b tagging requirements at WP 77%.

This baseline selection is applied to all events. The properties of the hadronically decaying W boson have not been exploited so far. In the next section, a procedure is derived for the explicit reconstruction of a W_{had} candidate.

6.2 On the reconstruction of W bosons

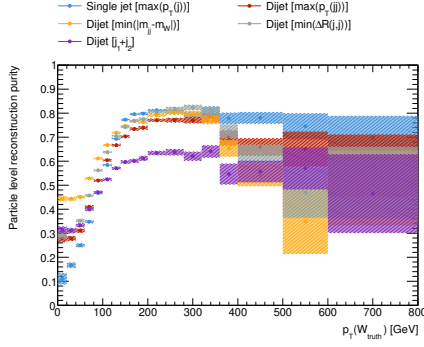
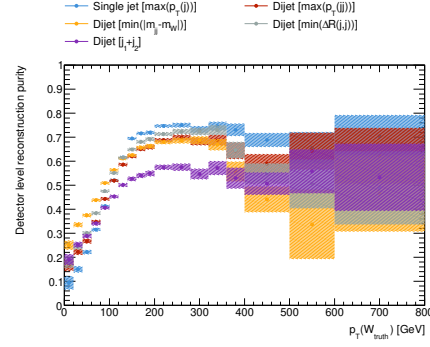
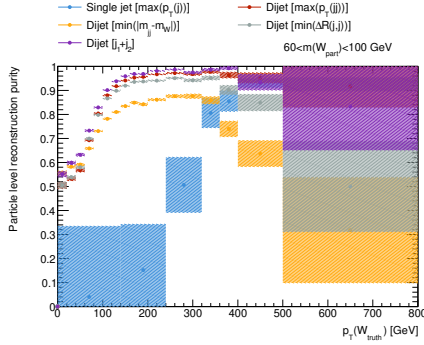
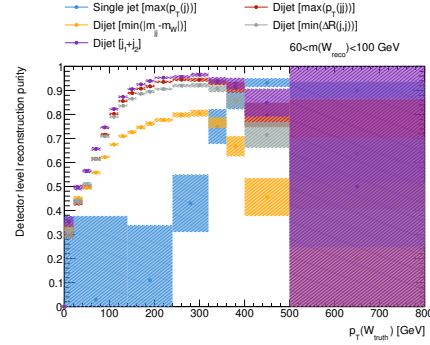
This section tackles the challenge of the reconstruction of the hadronically decaying W . While the W decays into exactly two quarks, the signature in the detector can look quite differently. In most cases, both decay products form jets which can be reconstructed with the standard jet clustering. There can also be a third jet, originating from a gluon radiated from one of the quarks. When the momentum of the W is sufficiently large, the system becomes collimated, or Lorentz-boosted. All decay products of the W are clustered into just one jet, since the two initial quarks can no longer be resolved. Thus, the difficulty in the reconstruction is two-fold, handling different event topologies, while solving the combinatorics arising from varying jet multiplicities. The goal of this section is to find one reconstruction algorithm that can solve the issues, while being easy to implement.

This study employs the the nominal $t\bar{t}$ sample with the loose selection described in the previous chapter. Different jet-combinations are considered as possible candidates for the W . The different choices can be compared by defining the purity p . It is defined as matched events divided by all events passing the loose selection. Matched events include all events where the angular distance ΔR between the true W before the decay W_{truth} with the candidate reconstructed from jets is smaller than 0.5:

$$p = \frac{\#(\text{Events with } \Delta R(W_{\text{truth}}, W_{\text{cand}}) < 0.5)}{\# \text{Events}} \quad (6.1)$$

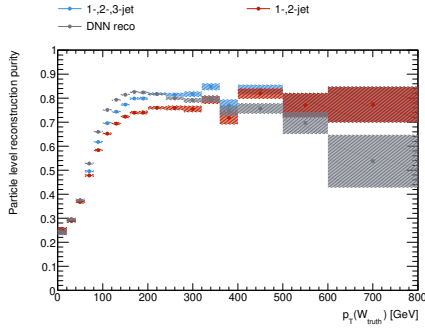
The jet combination that is selected is labeled as W_{cand} . We decided to focus on purity, instead of acceptance or efficiency, since the number of events in the analyzed data set is sufficiently large. Therefore, we can afford to sacrifice efficiency for the benefit of a cleaner event selection. The purity as a function of $p_T(W_{\text{truth}})$ for the comparison of W_{truth} to W_{particle} and W_{reco} is displayed in Figures 6.1(a) and 6.1(b). The error band depicts the statistical uncertainty of the MC sample. Various jet combinations are selected as the W candidate. The highest purities in the low p_T regime $p_T < 100$ GeV are obtained by selecting the dijet combination whose mass is closest to the known W mass (orange). The performance in the medium p_T range is similar for all considered options with a purity around 80%. Only the purple curve selecting the 4-vector sum of the leading and sub-leading light jet performs slightly poorer with a purity of around 60%. At high W momenta, a preference towards high-momenta candidates can be observed: The leading light jet (light blue) as well as the hardest dijet combination (red) perform best. The purities obtained for the particle and detector level studies are similar, with slightly larger values obtained for the particle level study.

The same plots are shown in Figure 6.4 with the addition of a selection cut on the mass of the W candidate $60 < m(W_{\text{cand}}) < 100$ GeV. This cut is applied in the nominator as well as the denominator of the purity definition. Single bins are merged in the plotting step to reduce the statistical uncertainties. Selecting the jet combination whose mass is closest to the true W mass, is not the best choice as W_{cand} (orange curve). The highest purity is obtained by selecting the dijet system containing leading and sub-leading light

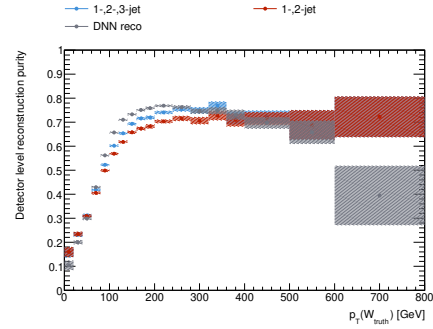
(a) Purity for the particle level W_{had} reconstruction.(b) Purity for the detector level W_{had} reconstruction.Figure 6.1: Comparison of the particle and detector level reconstruction of the hadronically decaying W from light jets.(a) Purity for the particle level W_{had} reconstruction.(b) Purity for the detector level W_{had} reconstruction.Figure 6.2: Comparison of the particle and detector level reconstruction of the hadronically decaying W from light jets. Only events passing $60 < m(W_{\text{cand}}) < 100$ GeV are considered for the calculation of the purity.

jet (purple). At highest W momenta, selecting the leading jet results in the highest purity. The statistical uncertainties are significantly smaller, compared to the other options. It becomes evident, that one should pre-select different jet combinations as preliminary candidates to account for the different topologies from resolved and boosted events.

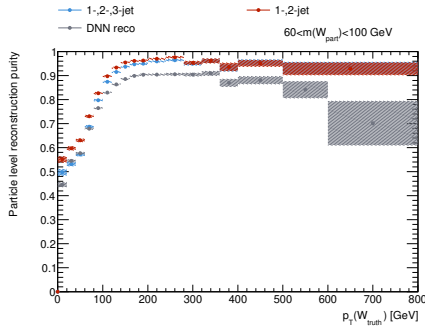
Figure 6.3 compares two different pre-selections. The blue curve labelled "1-2-3-jet" a total of five objects are pre-selected: the leading jet, the subleading jet, the vectorial sum of the two 4-vectors, the leading dijet and the leading 3-jet combination. The "1-2-jet" algorithm displayed in red drops the last two candidates from the list. While both options produce satisfactory results, the highest purity is obtained with the latter



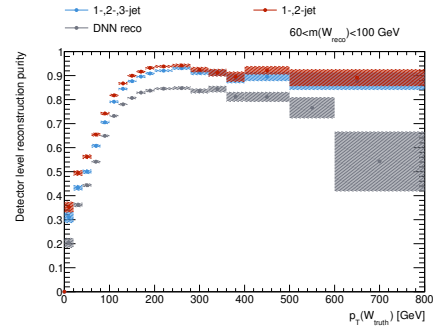
(a) Purity for the particle level W_{had} reconstruction.



(b) Purity for the detector level W_{had} reconstruction.



(c) Purity for the particle level W_{had} reconstruction with mass cut.



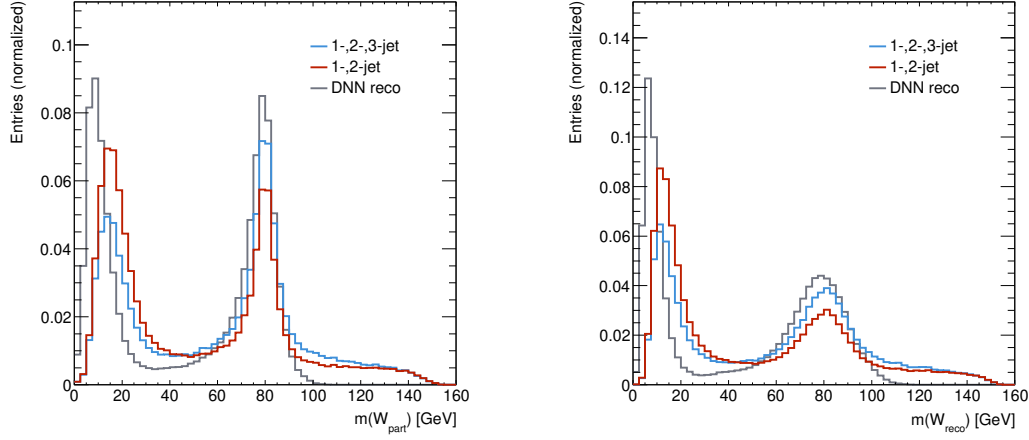
(d) Purity for the detector level W_{had} reconstruction with mass cut.

Figure 6.3: Comparison of the particle (left) and detector level (right) reconstruction of the hadronically decaying W from light jets. Different combinations of jets are pre-selected as the W candidate. The candidate that minimizes $|m(W) - m(W_{\text{cand}})|$ is selected as pseudo- W . Only events passing the $60 < m(W_{\text{cand}}) < 100$ GeV cut are considered in the lower row. The results are compared to the purities obtained from a DNN. More information is given in the text.

algorithm, when selecting a mass window around the W mass. This is the final algorithm that will be used in the analysis for event selection as well as observable definitions. The purity exceeds 90% for W momenta larger than 120 GeV.

Resolution plots are shown

A pseudo- W can be found for each event. Selection cuts can be applied on W_{cand} . The mass cut $60 < m(W_{\text{cand}}) < 100$ GeV is motivated by the purity studies. An additional cut on $p_T(W_{\text{had}})$ will be applied to ensure a good quality of the reconstruction for each event. A $p_T > 80$ GeV cut-off was chosen to ensure, that the purity exceeds 60% for the



(a) Mass distribution of the detector level W_{had} candidate. (b) Mass distribution of the detector level W_{had} candidate.

Figure 6.4: Comparison of the particle and detector level mass distributions.

particle and detector level reconstruction.

The results obtained from the classical reconstruction approach are compared to a Machine-Learning based reconstruction. A fully-connected classification neural network is designed and trained for that purpose. The five leading light jets are considered for that study. Every 1-, 2- and 3-jet combination built from the input jets is considered as a W candidate. This results in a total of 25 combinations. The 4-vectors of the jet combinations are passed as input to the network. The jet combination with the smallest distance to W_{truth} in ΔR is labeled with "1", while "0" is assigned to all other combinations as training label. The network consists of five layers with 8, 16, 32, 16, 8 nodes, respectively. This results in a total of 1401 trainable parameters. The layout of the neural network is illustrated in Figure....

The study is performed on $t\bar{t}$ samples generated with Powheg+Pythia. The data is split equally into training and validation samples with XXXX $t\bar{t}$ events each.

The purities are calculated on the validation sample. The statistical uncertainties are thus not comparable to the ones obtained in the classical method.

6.3 Measurement Regions

Final event selection and definition of the measurement phase space

Table with phase space cuts for loose and W_{had} region

Yield tables?

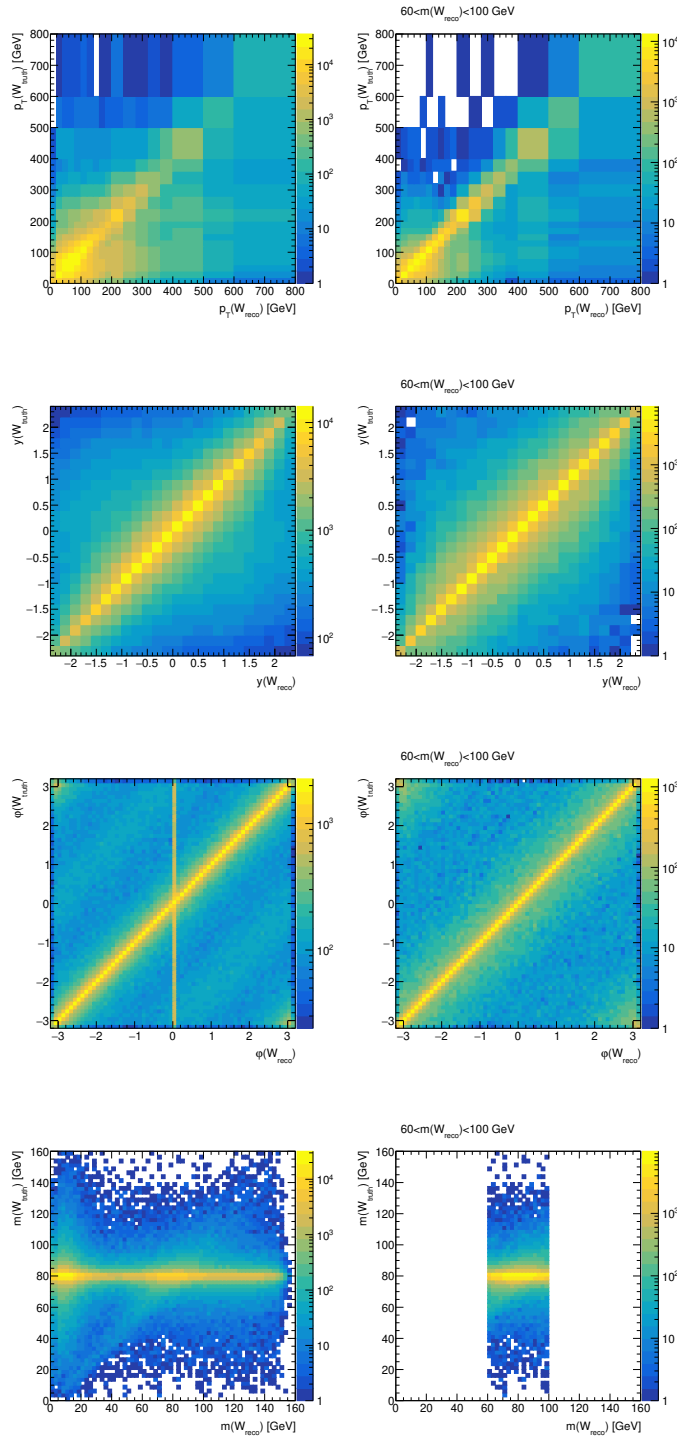


Figure 6.5: Add caption.

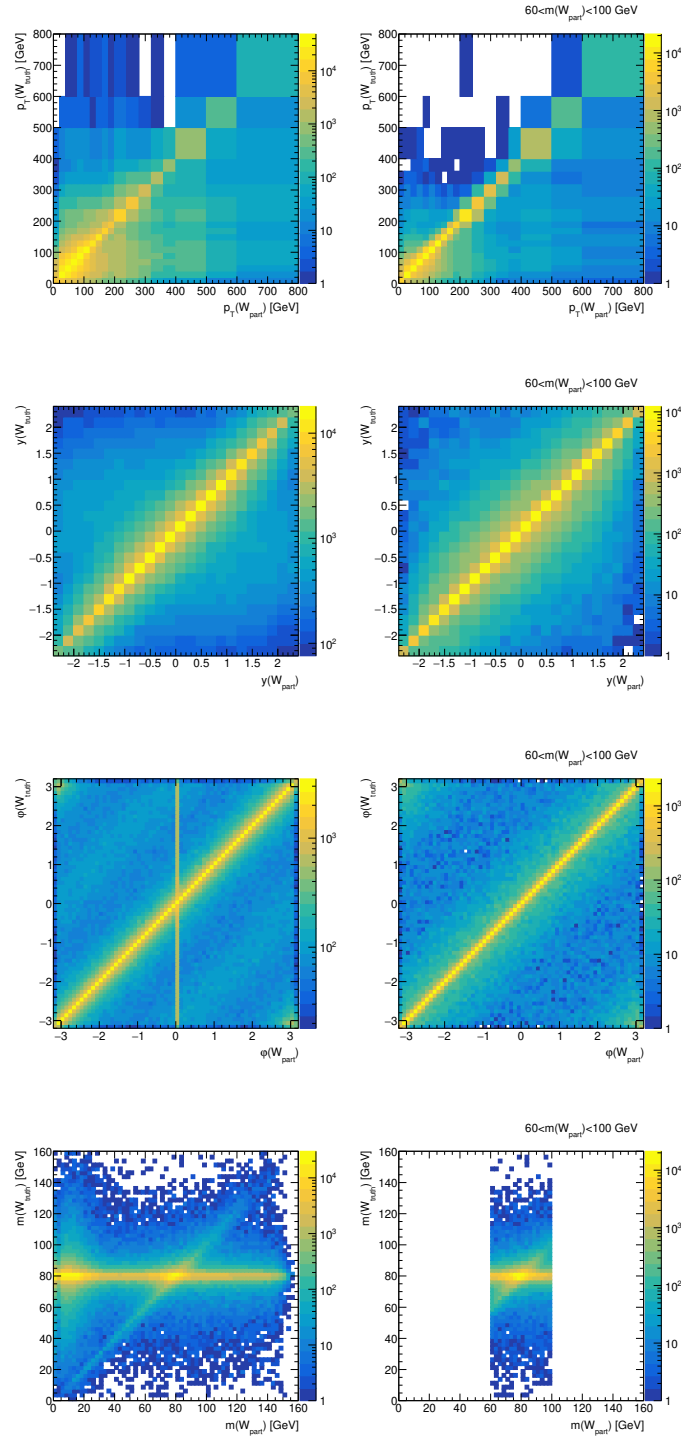


Figure 6.6: Add caption.

7 Extraction of the top quark mass

Introduce the idea.

7.1 The Linear Template Fit

Write a section on the LTF.

7.2 Templates for the fit

Production of the templates.

7.3 Results of the fit

Present competitive top mass results.

8 Conclusion

The measurement of the cross-section of several observables sensitive to the interference between $t\bar{t}$ and tW in the $WbWb$ final state is performed using the full Run 2 dataset collected by the ATLAS experiment.

This is the first measurement in the single lepton channel and its added value, with respect to the measurements in the dilepton channel, is probing new phase spaces with the aim of understanding the Monte Carlo modelling and reducing uncertainties for beyond the Standard Model searches. Furthermore, the semi-leptonic decay channel provides the opportunity to measure the hadronic W boson kinematics in the inclusive $WbWb$ final state.

The sensitivity of the analysis is currently limited by the statistics and by the uncertainty associated to the interference itself. In some distributions it is also masked by the top quark momentum mismodelling. Hence, further improvements can be achieved with the data collected in the Run 3 and by using new Monte Carlo samples that can be optimised with these measurements.

8.0.1 Machine learning based reconstruction of the hadronically decaying W boson

This section give a more in-depth description of the machine-learning model that is introduced for the reconstruction of the hadronically decaying W boson in Section...

This study demonstrates that the ML-based reconstruction performs well on the bulk of events, while struggling to capture the structure of the phase space edges. The five leading light jets are considered as input. All possible 1-, 2-, and 3-jet combinations are considered. A total of 25 jet candidates is found and their 4-vectors are passed to the network as input. A classification network is trained. The candidate which minimizes $\Delta R(W_{\text{truth}}, W_{\text{cand}})$ is labeled with "1", while all other candidates are assigned the label "0".

The candidate features are processed through a sequence of five dense layers with ReLU activation functions. The architecture consists of 8,16,32,16,8 nodes, respectively. A dropout layer with a dropout rate of 20% is added after the second and third layer to reduce the risk of overfitting during the training step. The 25 output nodes are normalized by applying a softmax activation function, such that the output score can be interpreted as a probability. The Adam optimizer with categorical cross-entropy is used as loss function.

Separate networks are trained for the particle and detector level reconstruction. The loss curves which are obtained for every single training epoch are displayed in Figure ??.

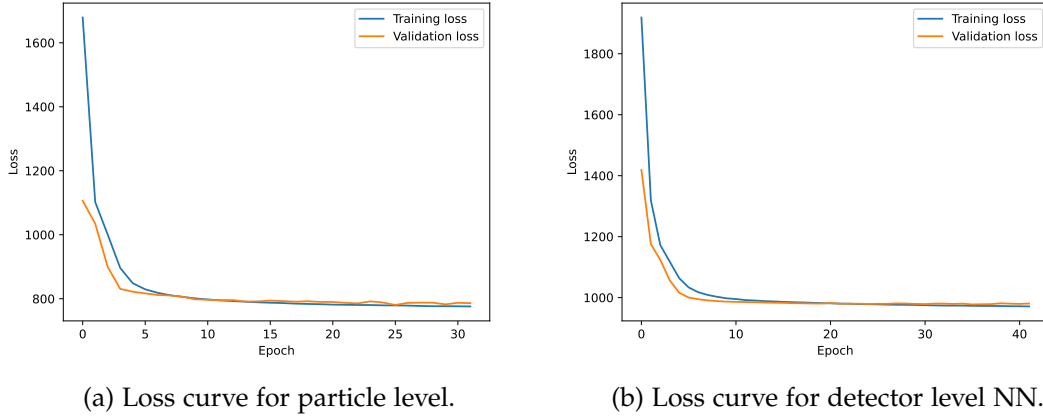


Figure 8.1: Loss function minimized during the training of the neural network. Separate networks are trained for the particle and detector level reconstruction. The loss curve is shown for the training as well as for the validation set.

The fraction of events used for validation is 20%.

After a steep decline, the loss is flattening out. Similar values are obtained for training and validation set, indicating no over-fitting of the model. Early stopping / values from best iteration used.

Figure 8.2 compares the particle level candidate identified by the NN to W_{truth} with respect to their kinematic properties. The resolution plots are dominated by the diagonal elements, with improving performance towards higher momenta. Similar results are obtained for the detector level reconstruction in Figure 8.3. The broadening of the diagonal elements is a direct consequence of the imperfect detector resolution.

Figures 8.4 and 8.5 show the index, W_{cand} mass and light jet multiplicity distributions for particle and detector level, respectively.

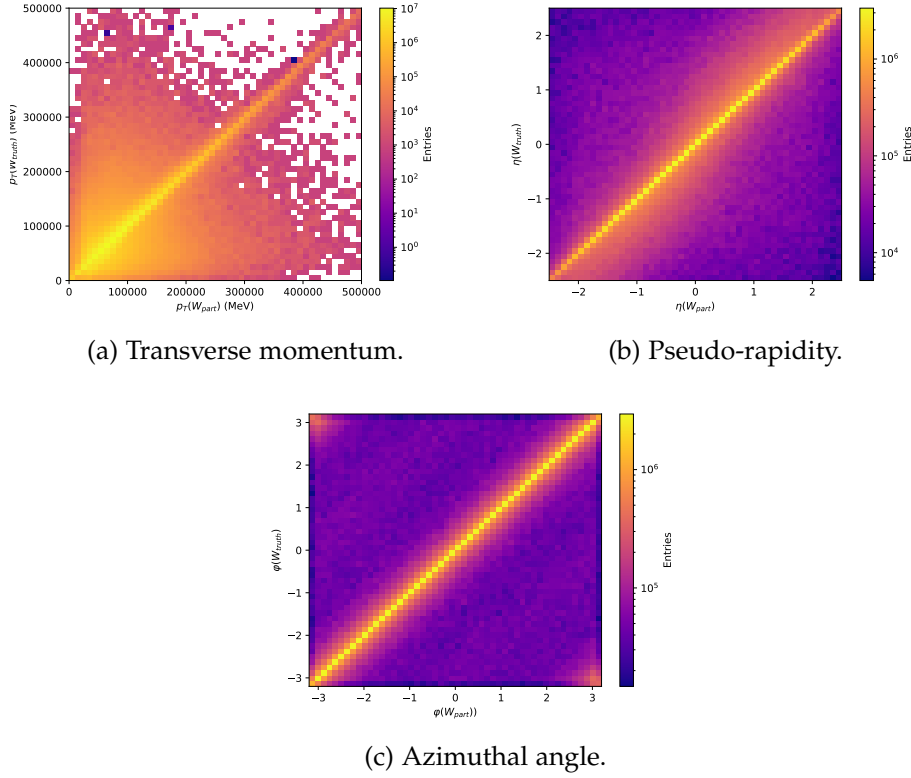


Figure 8.2: Resolution plots for the ML-based W_{had} particle level reconstruction. The kinematic observables transverse momentum p_T (8.2(a)), pseudo-rapidity η (8.2(b)) and azimuthal angle ϕ (8.2(c)) of the particle level W_{had} candidate are being compared to W_{truth} .

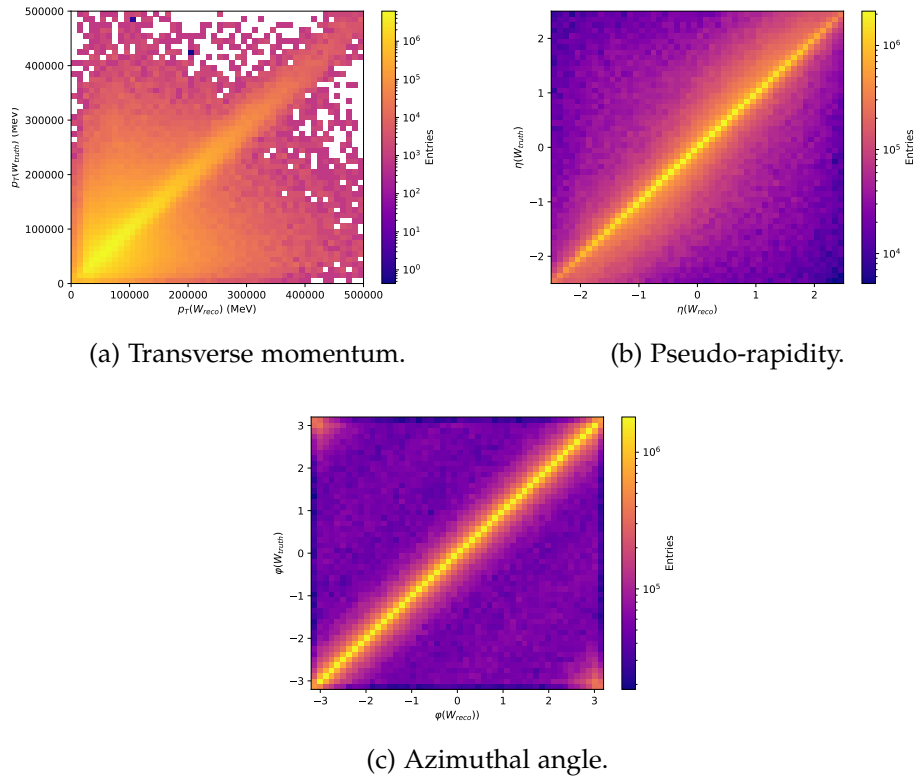


Figure 8.3: Resolution plots for the ML-based W_{had} reconstruction on detector level. The kinematic observables transverse momentum p_T (8.3(a)), pseudo-rapidity η (8.3(b)) and azimuthal angle φ (8.3(c)) of the W_{had} candidate are being compared to W_{truth} .

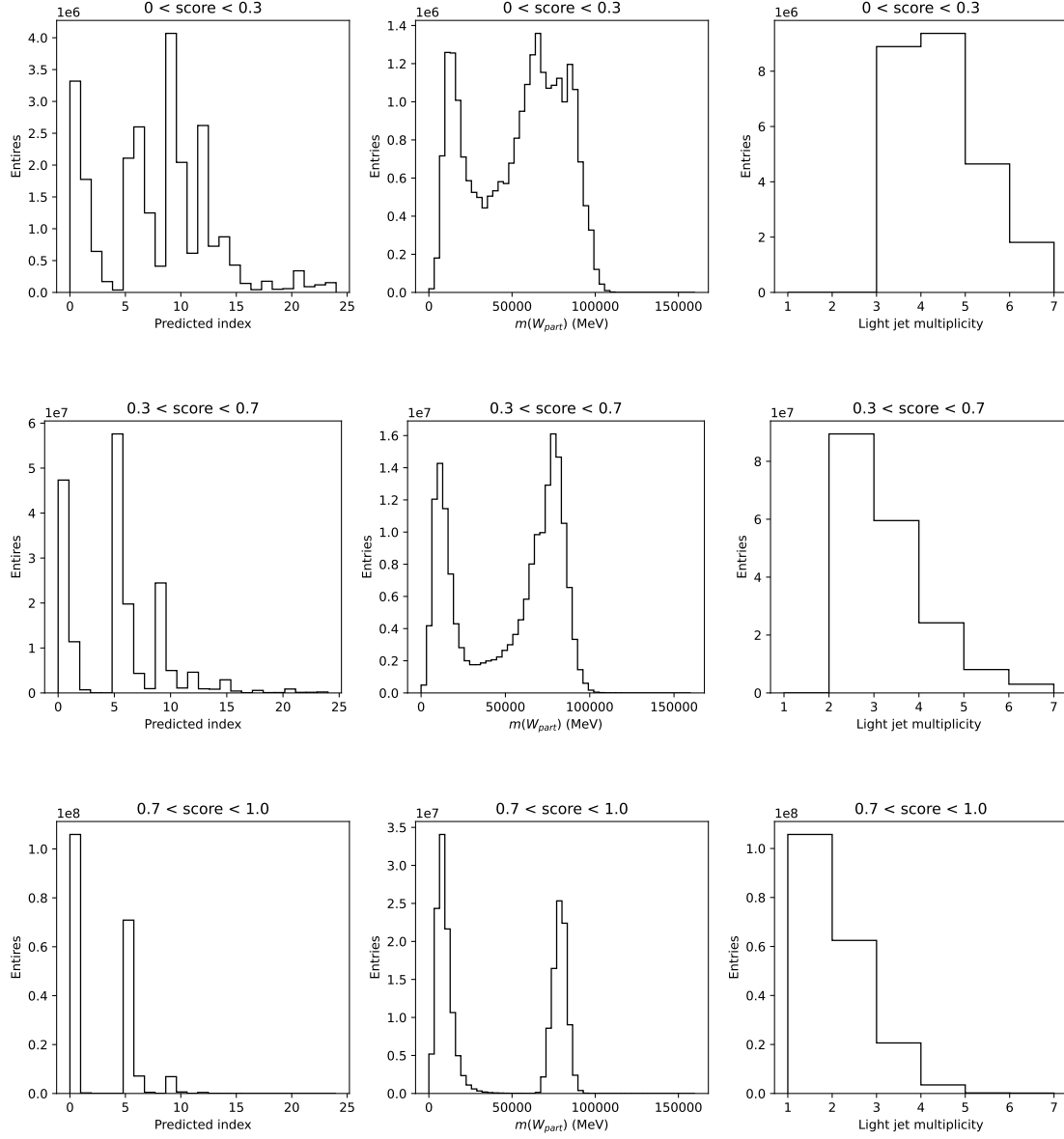


Figure 8.4: Loss function minimized during the training of the neural network. Separate networks are trained for the particle and detector level reconstruction. The loss curve is shown for the training as well as for the validation set.

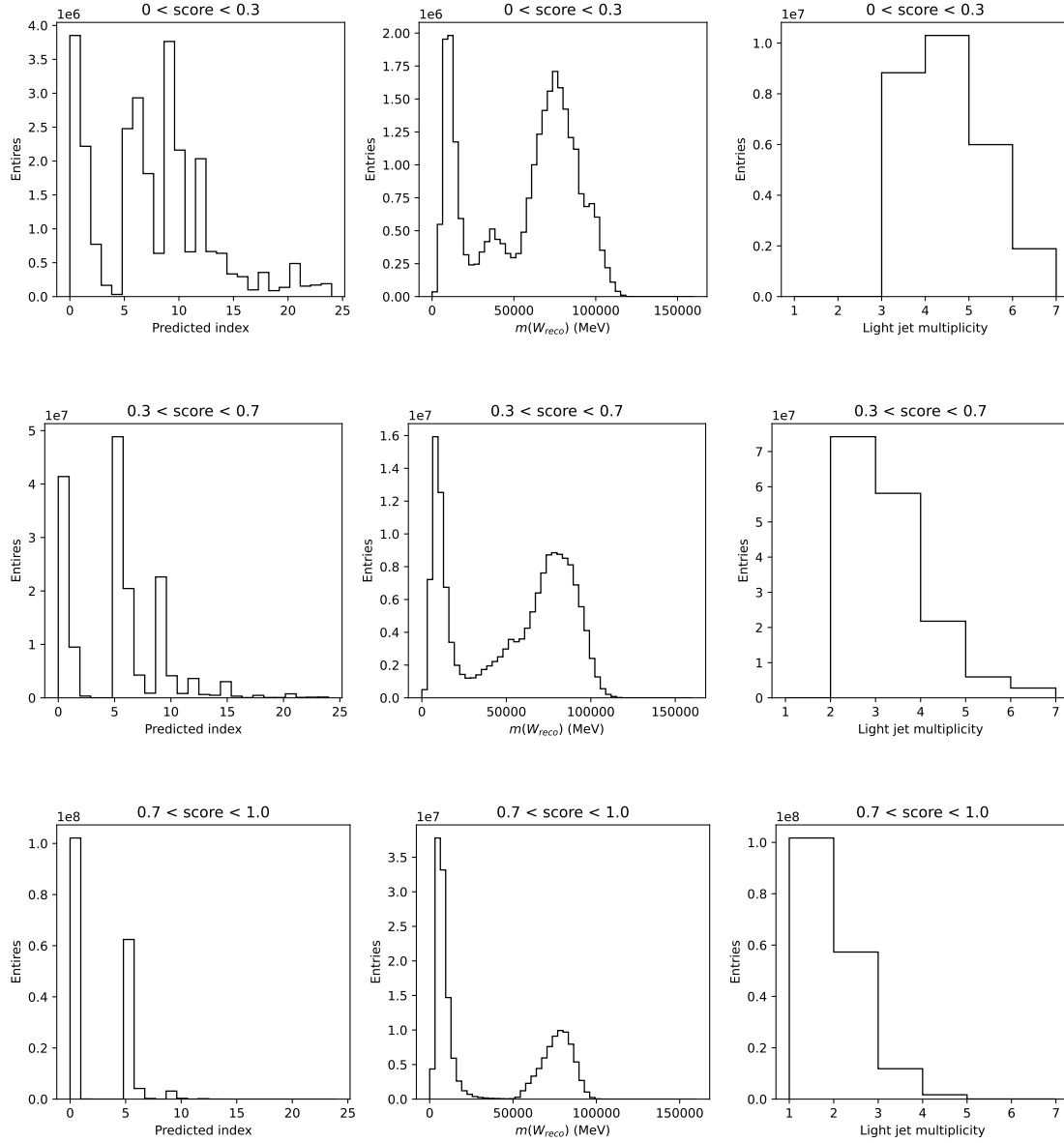


Figure 8.5: Loss function minimized during the training of the neural network. Separate networks are trained for the particle and detector level reconstruction. The loss curve is shown for the training as well as for the validation set.

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